Table 3 Taylor Backshell Outer Jacket Stripping Dimensions

Number of Pins on Connector	Dim 'A' for One Twisted Pair	Dim 'A' for Two Twisted Pairs
50/78	3.25 in (83 mm)	4.5 in (114 mm)
24/44	3.0 in (76 mm)	4.25 in (108 mm)
15/26	2.5 in (64 mm)	3.75 in (95 mm)
9/15	2.1 in (53 mm)	3.35 in (85 mm)

4) Trim the individual wires back to allow for more usable room inside the backshell.

5) Crimp on the terminals (pins or sockets) and insert into the correct location of the connector.

6) Place the Taylor backshell over the wire bundle.

7) Attach the connector to the backshell.

8) Insert the shield braid into the slot in the backshell. Do not allow any of the braids to cross over each other. (Refer to Figure 15)

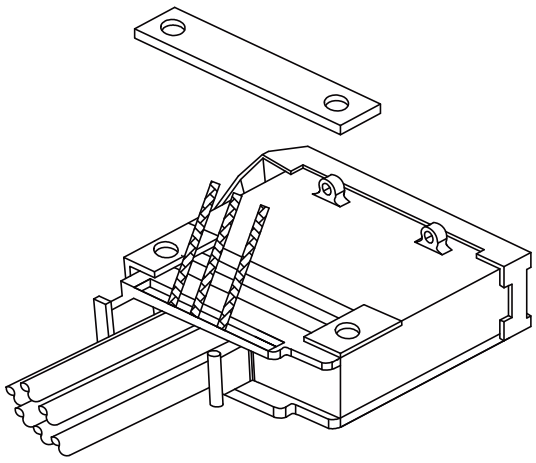


Figure 15

9) Clamp the braid to the backshell. Tightening the plate will press the braid against the backshell and provide a consistent electrical ground.

10) Trim any excess braid. (Refer to Figure 16)

11) Slide on the aperture cover until the cover completely closes the remaining opening in the shell.

Note Multiple sizes of this plate are available. Selection of the size is dependent on the bundle diameter and is the responsibility of the installer.

12) Tie wrap or lace the strain relief so that it holds the wires tightly. (Refer to Figure 16)

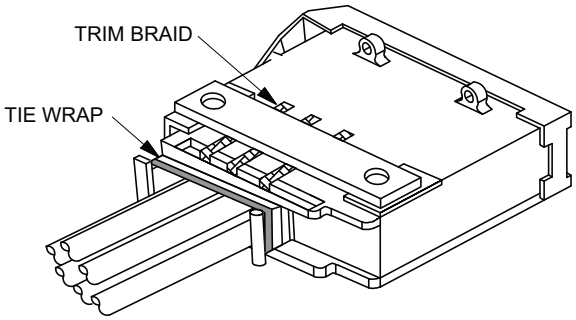
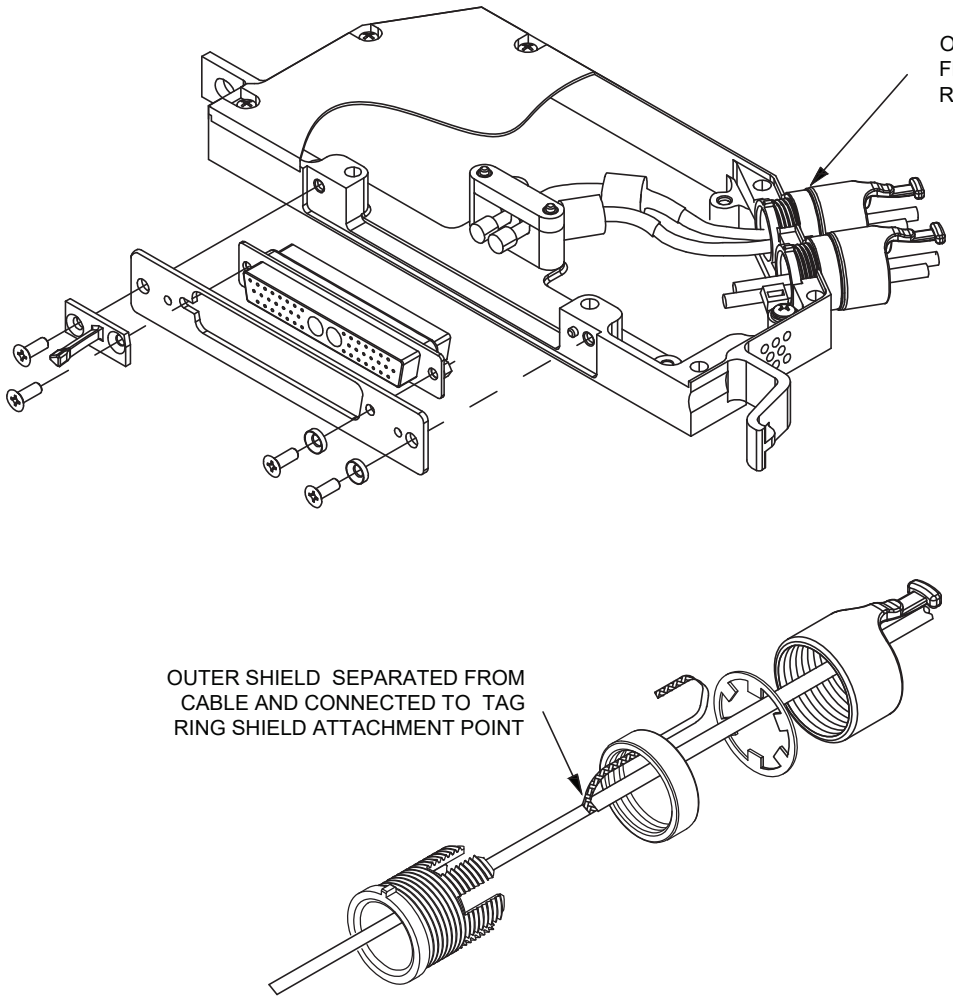


Figure 16

13) Ensure that the backshell is electrically bonded to the Line Replaceable Unit (LRU) according to bonding requirements specified in 20-30-10, Securing Connector Backshells.

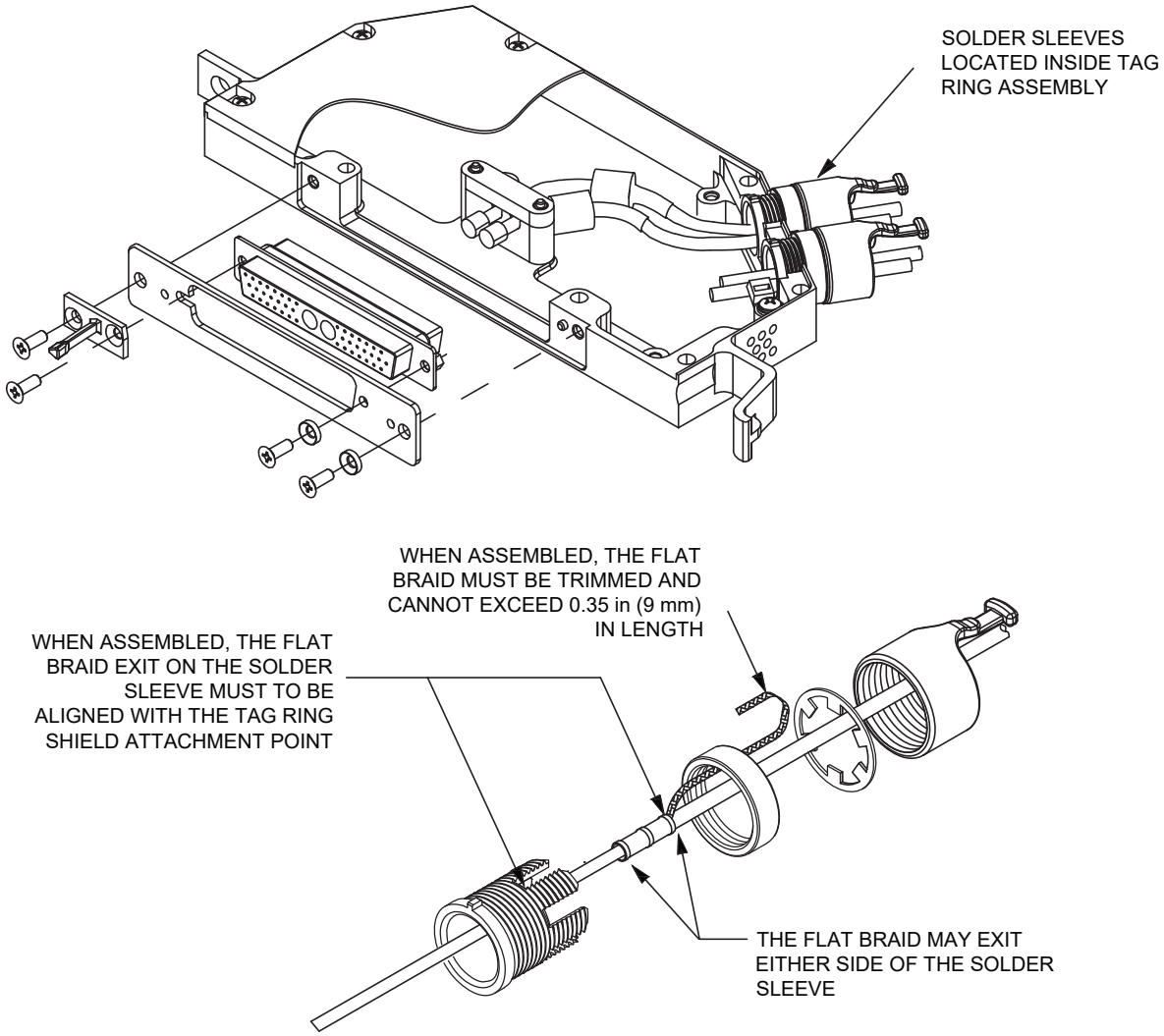
D. Terminating LAN Shields at NIC Backshells

1) For Pre Mod 6/2303, the outer shield of the LAN TRIAX cable is connected to the NIC backshell by separating the shield from the cable, and connecting it to the tag ring shield attachment point at the opening of the backshell. (Refer to)



LAN Shield Terminations at NIC Backshells (Pre Mod 6/2303)
Figure 17

2) For Post Mod 6/2303, the outer shield of the LAN TRIAX cable must be connected to the NIC backshell using a solder sleeve with flat braid (Raychem p/n SO63-4-9030). The solder sleeve must be installed at the opening of the backshell so that the flat braid exit on the solder sleeve aligns with the tag ring shield attachment point on the backshell (Refer to). The solder sleeve may be installed with the flat braid exiting either side of the sleeve. Once assembled, the flat braid must be trimmed and cannot exceed 0.35 inches (9 mm) in length.



LAN Shield Terminations at NIC Backshells (Post Mod 6/2303)
Figure 18

E. Terminating ASCB Shields at NIC Backshells

1) For Pre Mod 6/2303, the ASCB shield is not connected to the NIC backshell.

2) For Post Mod 6/2303, the ASCB shield must be connected to the NIC backshell in the same manner as the Post Mod 6/2303 procedure for connecting the LAN shields (Para 3.D.2)).

F. Terminating AFCS CAN Bus Shields at NIC Backshells (S.O.O. 6221)

1) The AFCS CAN bus shield must be connected to the AIOP backshell in the same manner as the Post Mod 6/2303 procedure for connecting the LAN shields (Para 3.D.2)).

G. Terminating ASCB Shields at FWD Maintenance Panel Connector Backshells (Post Mod 6/2303)

The following procedure describes the required method for attaching the ASCB shield to the Glenair 550-009 backshells mounted on the FWD maintenance panel. The backshells are installed Post Mod 6/2303.

Note The steps in this procedure are illustrated in Figure 19.

1) Remove 4 inches (102 mm) of jacket, shield and foil from the cable.

2) Remove an additional 0.75 inches (19 mm) of jacket to expose the cable shield.

3) Carefully comb out the shield.

4) Wrap the cable with 1 inch (25 mm) wide guideline tape (Freudenberg-NOK Inc. p/n 52672) to increase the overall diameter to approximately 0.45 inches (11 mm).

5) Carefully fold the shield back and overtop the guideline tape.

6) Maintain twist in the wires by installing a piece of 3/16 inch ID white heat shrink tubing overtop the exposed wires.

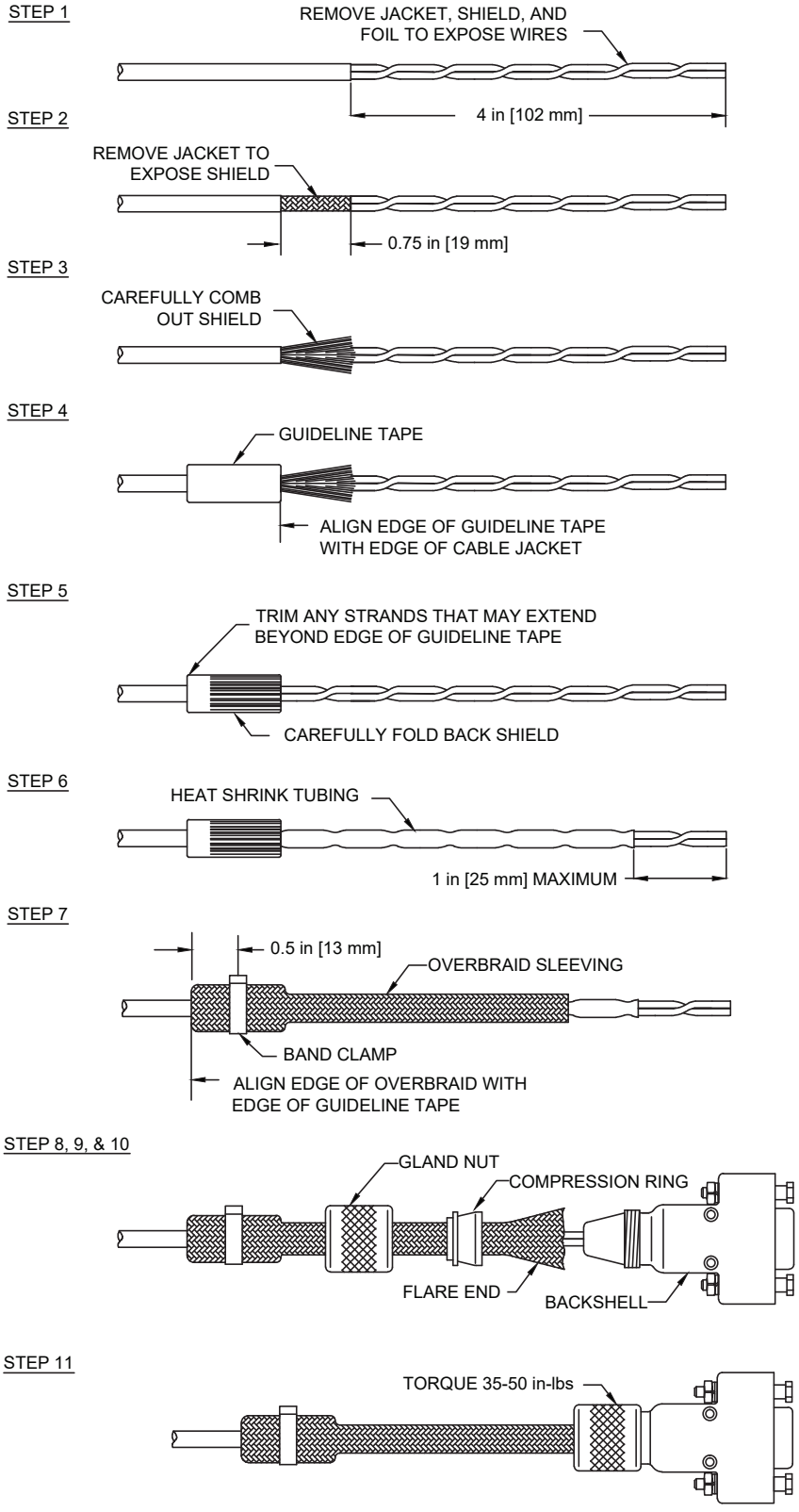
7) Cut a piece of overbraid sleeving (Raychem p/n Ray90-10) to a suitable length, and install the braid overtop the exposed wires. Fold 0.25 inches (6 mm) of braid into itself at the band clamp end, then install a band clamp (Band-it p/n AS2219) overtop the braid and guideline tape.

8) After crimping the connector contacts, slide the gland nut and compression ring overtop the cable.

9) Install the contacts into the connector, then install the backshell onto the connector.

10) Flare the overbraid, then slide the end over the tapered opening of the backshell.

11) Tighten the gland nut to 35-50 in-lbs to complete the connection.



Terminating ASCB Shields at Maintenance Connector Backshells (Post Mod 6/2303)
Figure 19

H. Terminating AFCS CAN Bus Shields at Servo Connector Backshells (S.O.O. 6221)

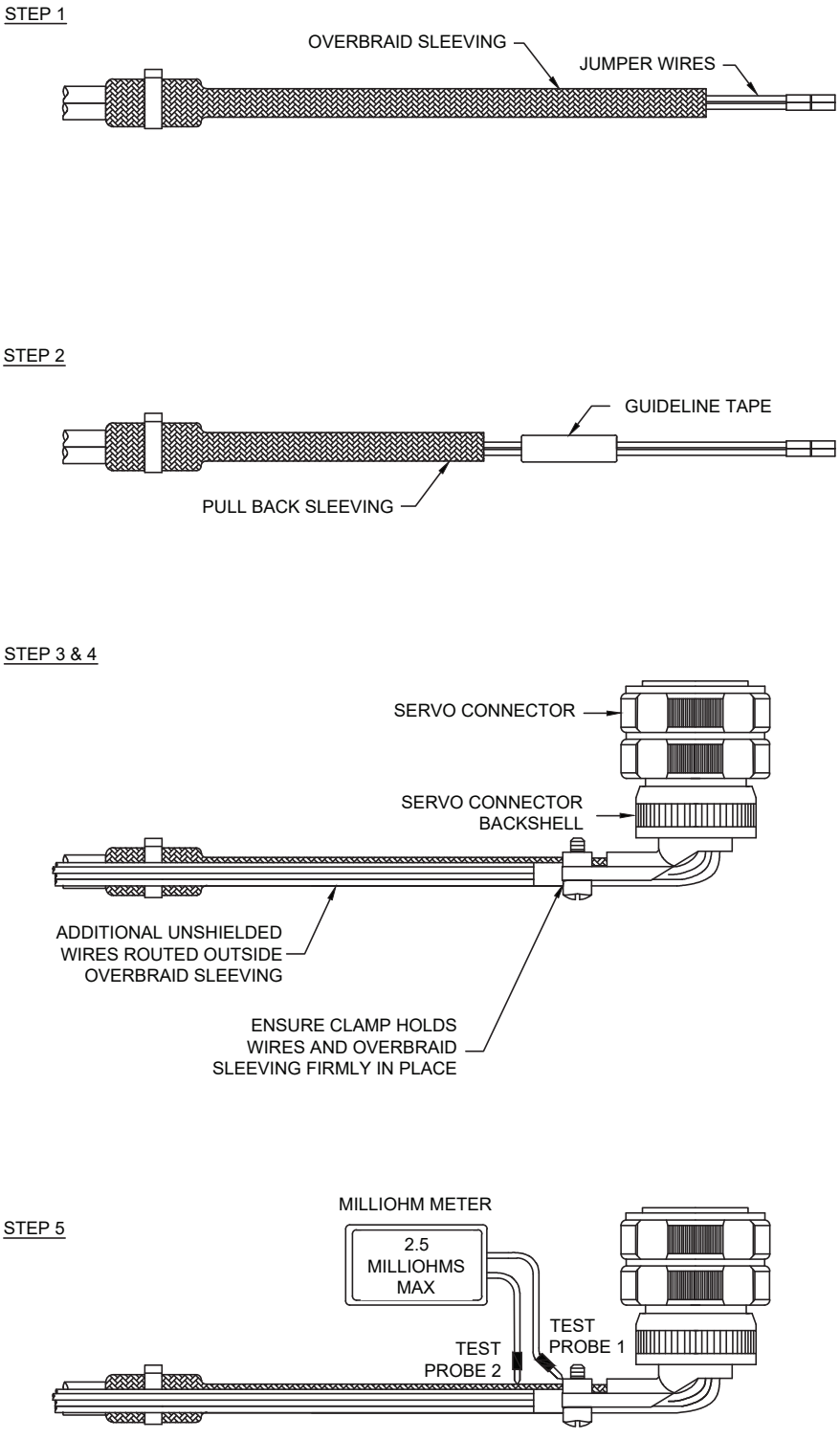
1) Attach overbraid sleeving to the AFCS CAN bus cable(s) as described in 20-10-60, Stub Construction, Para 3.C.

2) At the location where the overbraid sleeving will be secured underneath the backshell clamp, pull back the sleeving and wrap the wires with a single piece of guideline tape (Freudenberg-NOK Inc. p/n 52672). The guideline tape is used to increase the diameter of the bundle, which allows for better electrical contact between the overbraid sleeving and the backshell clamp.

3) Push the overbraid sleeving back overtop the wires and finish wiring the connector. For the unshielded wires routed outside of the overbraid sleeving, wrap the bundle with a single piece of guideline tape (Freudenberg-NOK Inc. p/n 52672) at the clamping point.

4) Tighten the backshell clamp. Ensure the clamp holds the bundle and overbraid sleeving firmly in place without crushing or straining the wires. If necessary, increase the amount of guideline tape to ensure the wires and overbraid are held firmly in place.

5) Measure the resistance between the overbraid sleeving and the connector backshell using a milliohm meter. The maximum allowable resistance is 2.5 milliohms. If the resistance is greater than 2.5 milliohms, check the bonding surface for contaminants and clean as required.



Terminating AFCS CAN Bus Shields at Servo Connector Backshells (S.O.O. 6221)
Figure 20

Shield Terminations Using HexaShields

1. General

This section describes shield termination procedures for Raychem HET-A-02X, HET-A-03X, and HET-A-04X HexaShield termination kits.

HET-A-02X and HET-A-04X HexaShields are used to terminate braided wire shields with the use of a ferrule instead of a solder sleeve. Each kit consists of one plated metal ferrule, one SolderShield heat-shrinkable solder device, and one short piece of heat shrinkable tubing. Once installed, the SolderShield forms an electrical connection between the ferrule and shield. The ferrule is then clamped to the backshell to provide a reliable ground path for the shield. (Refer to Figure 2 and Figure 3)

HET-A-03X HexaShields are used as a feed-through for unshielded wire when shield grounding is not required.

Hexashields are indicated with the following wiring symbols:

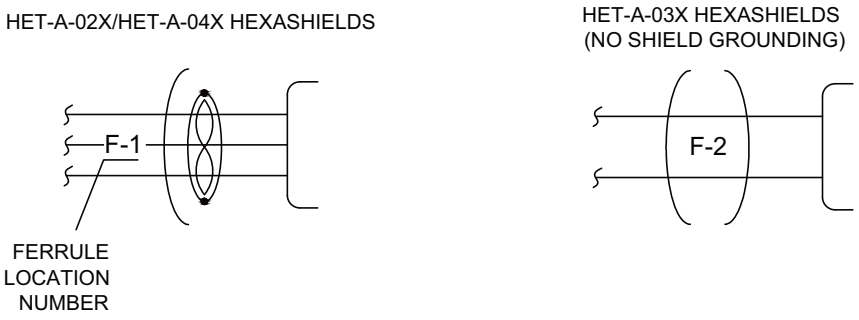
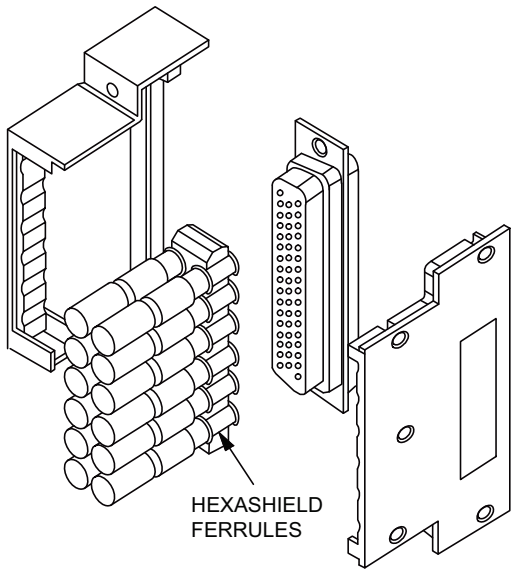
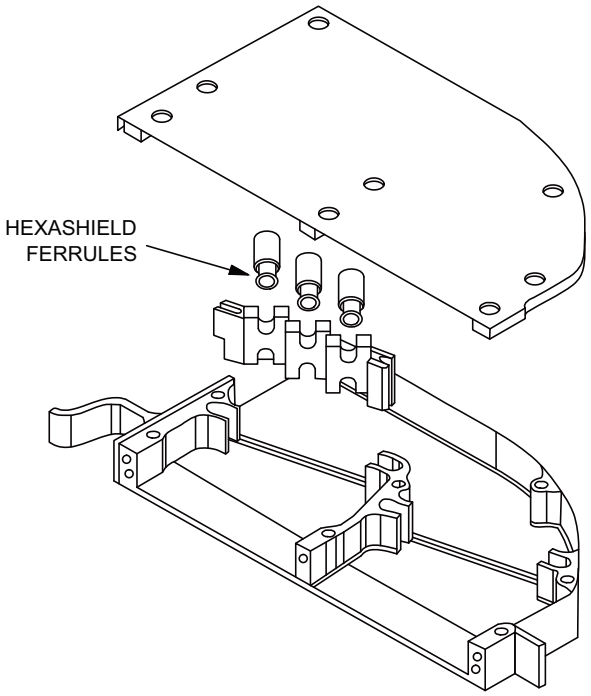


Figure 1

Applications include shield grounding at the Air Data, Attitude, and Heading Reference System (ADAHRS) and Advanced Graphics Module (AGM) backshells.



ADAHRS Backshell
Figure 2



AGM Backshell
Figure 3

2. Materials and Equipment

A. Materials

1) HexaShield installation kits, Raychem p/n HET-A-02X, HET-A-03X, HET-A-04X

2) HexaShield dummy plug, Raychem p/n HET07-AC

3) Solid, uninsulated, tin plated copper wire, ø0.008 to 0.016 inches (ø0.2 to 0.4 mm)

B. Equipment

1) Ferrule support fixture, Raychem p/n ECE-0188

2) Holding fixture, Raychem p/n AD-1319–9

3) Wire and cable strippers, commercially available

4) Wire cutters, commercially available

5) Recommended hot air tools:

a) Raychem CV-1981 with PR-25D reflector, temperature control set to 7.5/8.5 with vent open (420 °C ± 20 °C)

b) Steinel HL 1802 E with PR-25D reflector and PR adapter, or Steinel 5/8” reflector, temperature control set to 11/12, high fan speed

3. Procedures

A. Installation of HexaShields

The following procedure describes the standard method for installing HET-A-02X and HET-A-04X shield termination kits. Similar procedures can be followed for installing HET-A-03X for unshielded wire.

1) For each shielded cable, strip the outer jacket and braided shield back based on the dimensions in Figure 4.

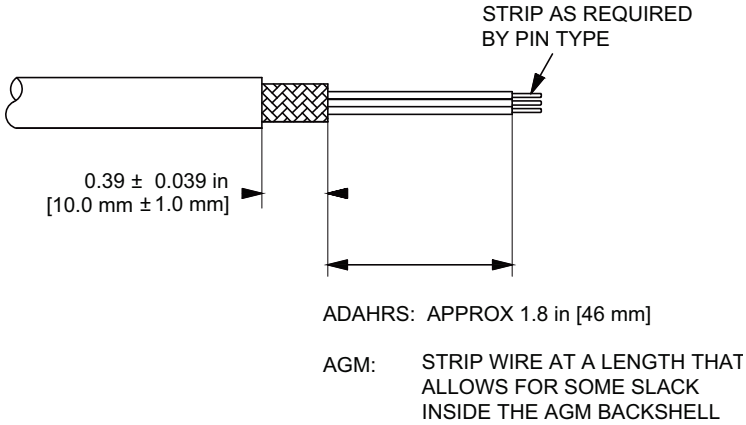


Figure 4

2) Determine the size of HexaShield to use based on the application:

a) ADAHRS backshells: HET-A-04X

b) AGM backshells: HET-A-04X

Note Up to four shields can be terminated with one ferrule. In order to avoid overfilling a ferrule, follow the ferrule assignments outlined on each wiring diagram.

Note Fill all unused ferrule locations with HET07-AC dummy plugs

3) Install the ferrule support ECE-0188 on the holding fixture AD-1319–9. Adjust the distance between the ferrule support and the cable clamp. Align the V shaped groove in the ferrule support with the V groove in the cable clamp. Tighten the Allen head screws to secure the support and clamp to the folding fixture. (Refer to Figure 7).

4) Optional

If four shielded cables are used in one ferrule, tie them together as shown in Figure 5, using tin plated copper wire. Trim ends to 0.20 inches (5.0 mm) long and fold flat along length of cable.

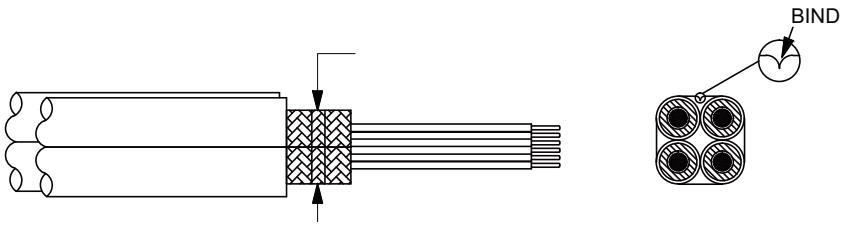


Figure 5

5) Slide the short piece of heat shrinkable tubing provided with the installation kit over the primary wires until the tubing butts against the cable braid(s). This tubing must be used with all primary wires less than 0.04 inches (1.0 mm) diameter to prevent small gauge wires from being damaged by the ferrule slot. The tubing secures and aligns multiple cables during termination and its use is recommended for all primaries. Heat the tubing in place.

6) Slide the SolderShield device and the ferrule over the cable as shown in Figure 6.

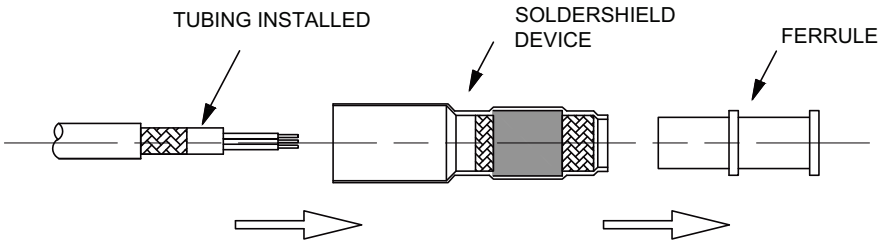


Figure 6

7) Position the assembly in the holding fixture according to the dimensions shown in Figure 7. The ferrule is correctly positioned in the V groove of the ECE-0188 adapter when the two larger diameter flanges of the ferrule rest on either side of the adapter V groove.

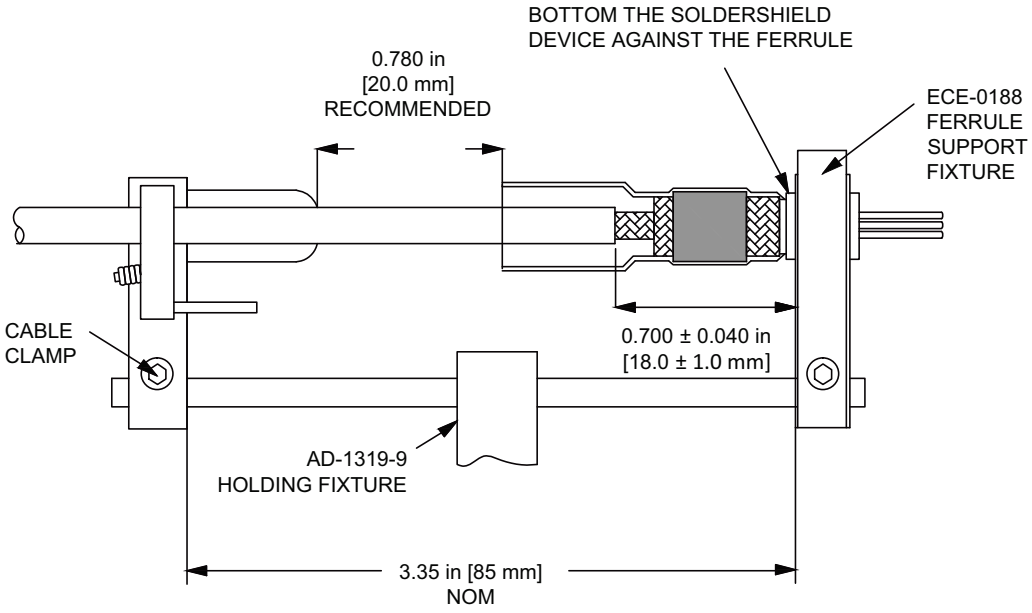


Figure 7

8) Allow the hot air tool (refer to Para 2.B for recommended equipment) to warm up for at least one minute to reach its operating temperature before heating the assembly.

9) Position the solder preform in the center of the reflector.

10) Heat until the solder preforms are completely melted and solder fillets are formed between the ferrule and SholderShield device, and between the cable braids and SolderShield device. If necessary, continue to heat the device, moving towards the cable until the tubing is fully shrunk on to the cable jacket(s).

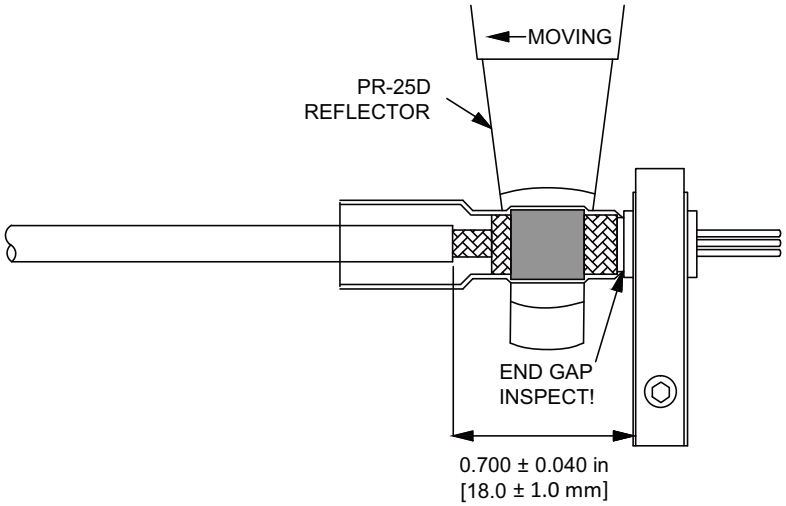


Figure 8

11) Allow the assembly to cool before removing it from the holding fixture.

12) Inspect the installation:

a) The 0.700 ± 0.040 inch (18.0 ± 1.0 mm) dimension must be maintained.

b) The SolderShield device braid must not overlap or touch the cable jacket.

c) The solder preform must have completely melted and flowed along the braid(s).

d) Visible remnants of the original shape of the solder preform indicate an underheated termination.

e) A highly discolored dark brown sleeve indicates an overheated termination. Slight browning of a sleeve is acceptable.

f) The gap between the end of the sleeve and shoulder of the ferrule must not exceed 0.051 inches (1.3 mm).

g) Inspection for damage:

1 The insulating sleeve must not be cut or split.

2 There must be no braid strands poking through the sleeve.

3 The insulation sleeve and cable jacket(s) must not show signs of mechanical damage or overheating such as cuts, melting, or charring.

B. Repair of HexaShields

1) Underheated Terminations

Reheat underheated termination to obtain proper solder flow and/or fully recovered sleeve onto the cable jacket.

2) Overheated Terminations

Refer to Para 3.C.

C. Removal of HexaShields

1) Score the insulation sleeve circumferential, approximately 0.28 in (7.0 mm) from the lip of the ferrule. Score the insulation sleeve lengthwise and remove the insulation sleeve. Heat may be used to assist in sleeve removal. Be careful not to cut the cable jacket.

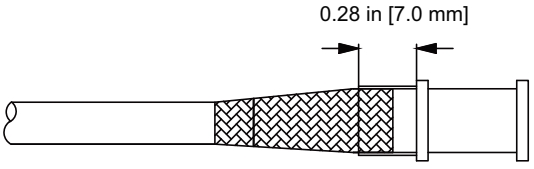


Figure 9

2) Grip the ferrule with pliers. Heat the SolderShield device to the cable braid solder joint until the solder melts, then remove the ferrule and SolderShield device by pulling it straight off the cable with the pliers. If the ferrule and braid do not come off together, thoroughly heat the braid and gently pull it off the cable. The solder must be fully melted or the cable braid may be damaged, particularly flat braids.

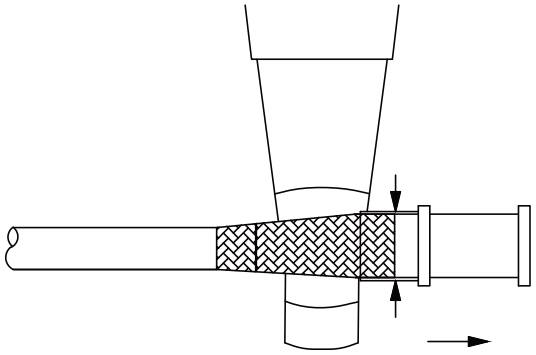


Figure 10

3) If it is necessary to separate multiple cables that were used through a single ferrule, remove the shrinkable tubing over the primaries (if present). Heat the braids until the solder melts. Separate the cable shields by gently pulling them apart.

This Page Intentionally Left Blank

Securing Connector Backshells

1. General

This section describes procedures for securing backshells to connectors. Correct installation prevents the backshell from unintentionally loosening and when applicable, provides proper bonding to maintain effective cable shielding. Procedures and requirements in this section must be followed unless specified otherwise.

2. Materials and Equipment

A. Materials

1) Tamper proof sealant, Organic Products Co. p/n F-900 or equivalent

B. Equipment

1) Recommended scale torque wrenches:

a) Daniels BT-BS-601

b) Glenair TG70

2) Soft jawed pliers, commercially available

3) Strap wrench, commercially available

4) Milliohmmeter, BCD M1 or equivalent

3. Requirements

A. All wire insulation, braided shields, and outer jackets must be checked for damage prior to termination.

B. Only Teflon tape shall be used on conductive surfaces. All tape shall be removed from conductive surfaces before completion of the assembly.

C. When threaded backshells are being secured:

1) The backshell must be screwed onto the connector using an actual torque of 40 in-lbs ± 1 in-lbs. (Refer to Para 4.A for procedure)

Anti-rotation teeth shall be locked in place, when applicable.

D. There must be no evidence of looseness of any terminating hardware.

E. Neither the connector nor the backshell shall have been deformed during the installation process.

F. Terminating hardware must not be deformed.

G. When terminating shields at the backshell:

1) Cable shields must be terminated using the practices specified in Para 1.

2) Proper electrical bonding is required for all backshells with shield terminations.

- For bulkhead mounted connectors, the resistance measured between the backshell and the connector must be less than 0.1 ohms. (Refer to Figure 1)
- For Line Replaceable Unit (LRU) connectors, the resistance measured between the backshell and the LRU mounting tray (or closest airframe bonding point on the LRU) must be less than 0.1 ohms. (Refer to Figure 1)

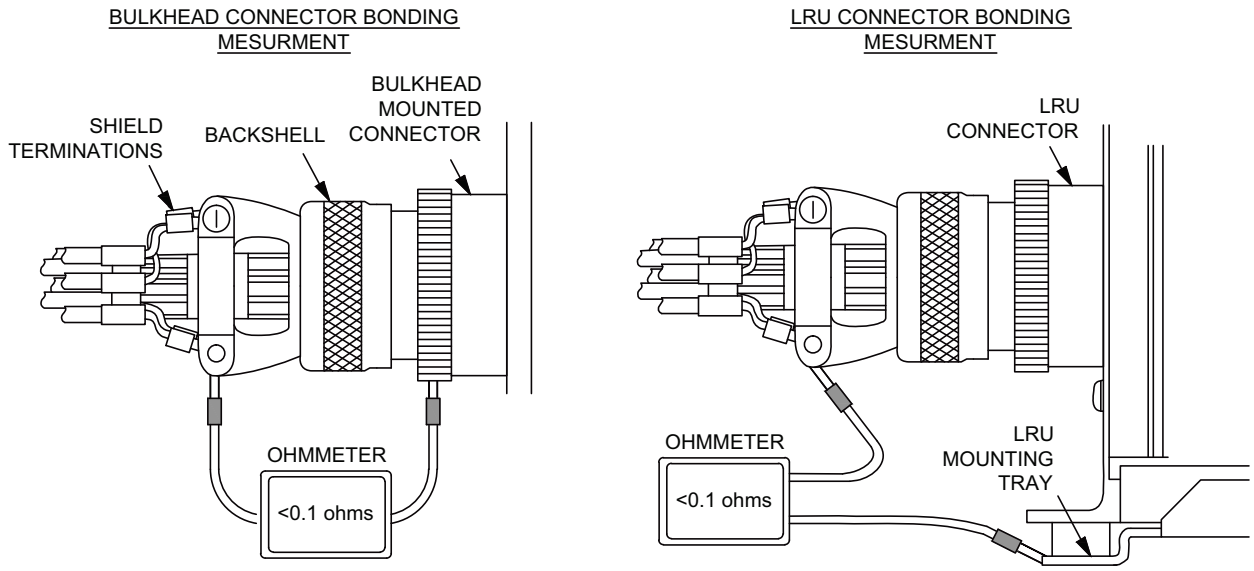


Figure 1

4. Procedures

A. Securing Threaded Backshells

WARNING DO NOT SECURE BACKSHELLS USING ANY KIND OF ADHESIVE OR SEALANT AS THIS COULD HAVE A DETRIMENTAL EFFECT ON ELECTROMAGNETIC INTERFERENCE (EMI) AND HIGH INTENSITY RADIO FREQUENCY (HIRF) PROTECTION OF THE WIRE HARNESS.

1) Remove any original torque stripe marks before securing the backshell.

2) Use a torque wrench fitted into a strap wrench to tighten the backshell onto the connector to the value specified in Table 1 — Corrected Torque Values.

Note Dial and click type torque wrenches may also be used.

Table 1 Corrected Torque Values

CONNECTOR/ADAPTOR DIAMETER	CORRECTED TORQUE (± 1 in-lbs)	
	DANIELS BT-BS-601 WRENCH	GLENAIR TG70 WRENCH
Up to 0.875 inch (22 mm)	35 in-lbs	30 in-lbs
0.875 - 1.25 inch (22 - 32 mm)	34 in-lbs	
1.25 - 1.875 inch (32 - 48 mm)	33 in-lbs	
1.875 - 2.125 inch (48 - 54 mm)	32.5 in-lbs	
2.125 inch - 2.375 inch (54 - 60 mm)	32 in-lbs	

a) Insert the square peg of the torque wrench into the square hole in the strap wrench so that the handles of the two wrenches are in line with one another. (Refer to Figure 2)

Use a torque wrench of such capacity that the required torque is within the middle two quarters of the scale (e.g. the best working range for a torque wrench with a 0–60 inch-lbs capacity is 15 to 45 pounds).

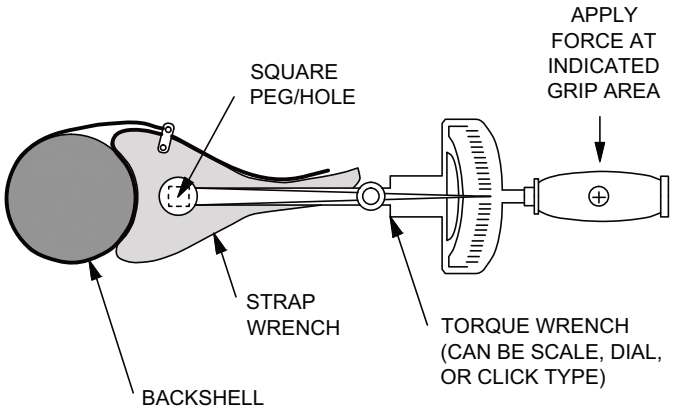


Figure 2

b) Hold the connector in place using one of the following means:

- Soft jawed pliers
- Strap wrench
- Secure the mate of the connector to a stationary object such as a board and hold the connector in place by inserting it into its stationary mate.

c) Grip the torque wrench by the pivoting handle or the grip area indicated in Figure 2 (usually marked by a knurl). Gripping the torque wrench anywhere other than by the handle or grip area will alter the effective lever length, resulting in an inaccurate torque reading.

d) Apply slow, steady pressure.

1 Keep the lay of the wrench parallel to the face of the connector.

2 Apply the torquing force at approximately 90° to the handle of the torque wrench.

e) For scale type torque wrenches, read the torque indicating dials and scales from directly above the indicator. Reading the indicator from an angle will result in an inaccurate torque reading.

f) After torquing, apply a torque stripe mark of tamper proof sealant onto the connector body overlapping onto the backshell. The mark must be a continuous stripe approximately 0.125 inches (3 mm) wide.

This Page Intentionally Left Blank

CHAPTER 21

AIR CONDITIONING

This Page Intentionally Left Blank

CHAPTER 21 – AIR CONDITIONING

This list shows the effective pages in the manual, with their latest revision date. An asterisk (*) indicates pages added or replaced at the latest revision. Technical changes to the content are indicated by a black vertical line next to the change or the paragraph that contains the change.

REFERENCE	PAGE	DATE	REFERENCE	PAGE	DATE
21	1	Sep. 30/14	21-40-00	1	Sep. 30/14
	2	Blank Page		2	Blank Page
	* 3	Nov. 30/18	21-50-00	1	Sep. 30/14
	* 4	Blank Page		2	Blank Page
	* 5	Nov. 30/18	21-56-00	* 1	Sep. 30/14
	* 6	Blank Page		* 2	Blank Page
21-21-00	1	Sep. 30/14	21-60-00	1	Sep. 30/14
	2	Blank Page		2	Blank Page
21-22-00	1	Sep. 30/14			
	2	Blank Page			

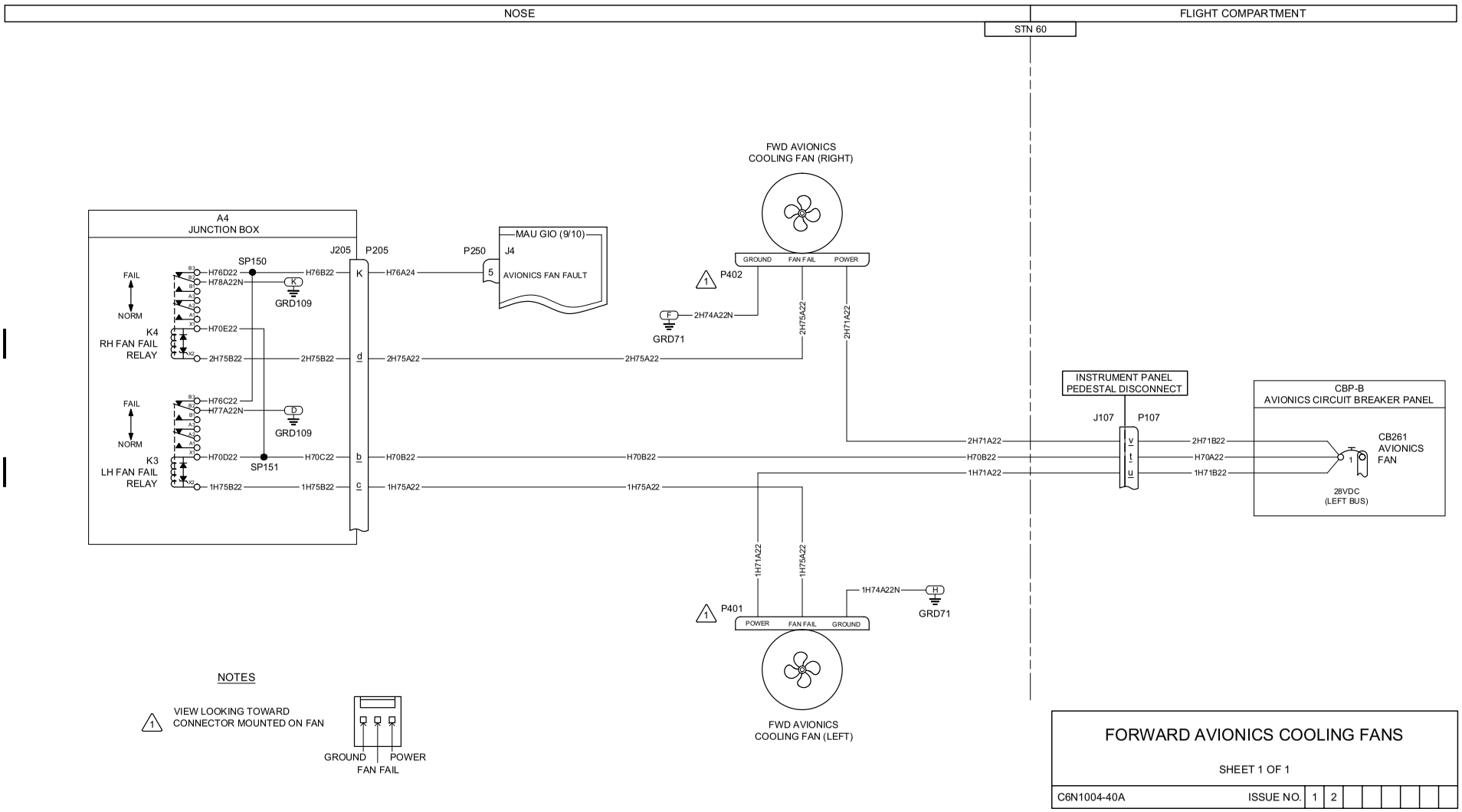
This Page Intentionally Left Blank

CHAPTER 21 – AIR CONDITIONING

TABLE OF CONTENTS

SUBJECT	CHAPTER SECTION SUBJECT
DISTRIBUTION	**21-20**
– FORWARD AVIONICS SHELVES	**21-21**
Forward Avionics Cooling Fans	21-21-00
– DATA ACQUISITION UNIT (DAU)	**21-22**
DAU Cooling Fans	21-22-00
HEATING	**21-40**
Bleed Air System	21-40-00
COOLING	**21-50**
Fan, Cooling & Demisting (S.O.O. 6322)	21-50-00
– CABIN FANS	**21-56**
Cabin Fans (S.O.O. 6378 & S.O.O. 6379)	21-56-00
TEMPERATURE CONTROL	**21-60**
Temperature Indication and Control System	21-60-00

This Page Intentionally Left Blank

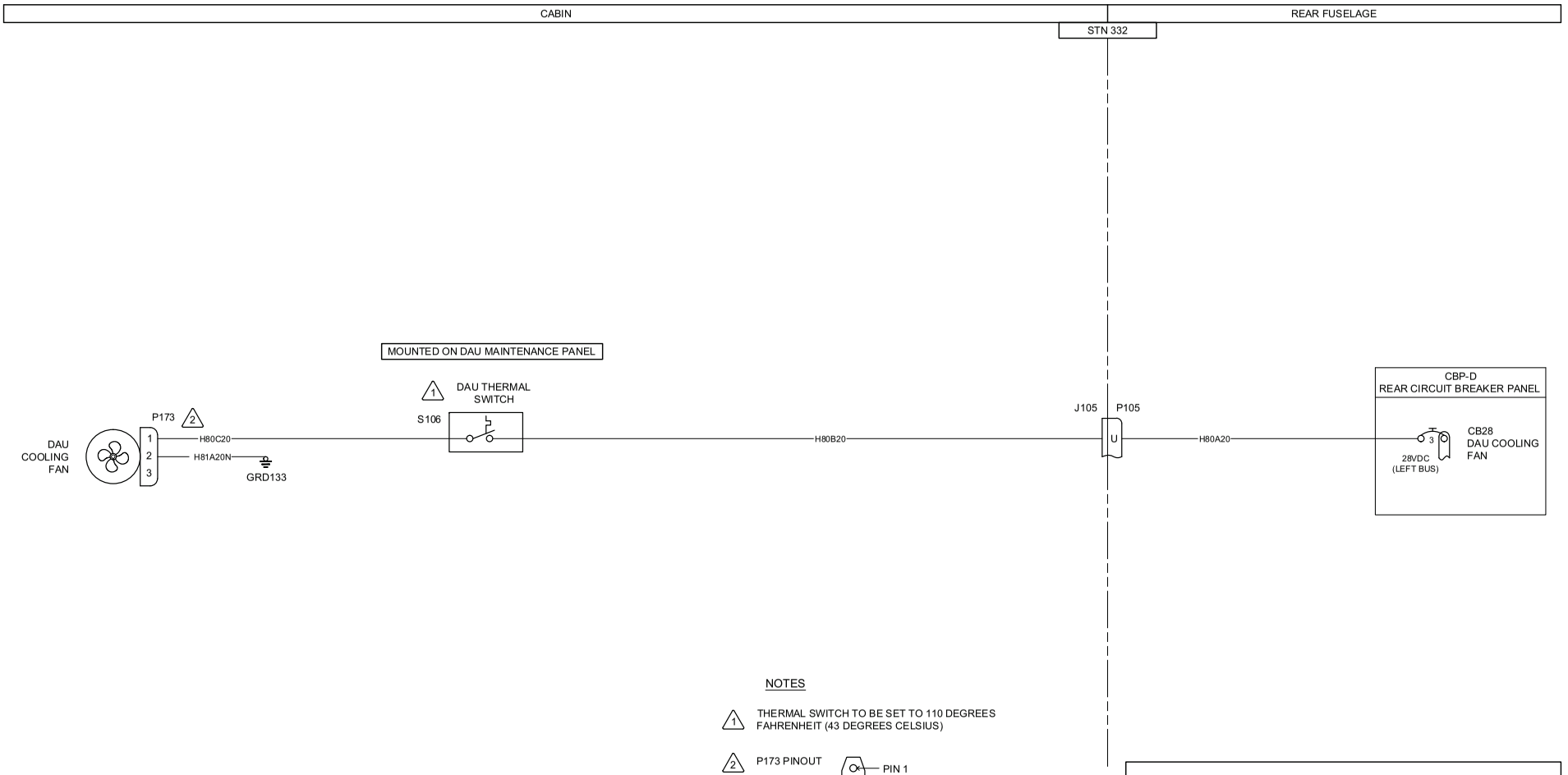


FORWARD AVIONICS COOLING FANS

SHEET 1 OF 1

C6N1004-40A	ISSUE NO.	1	2						

This Page Intentionally Left Blank

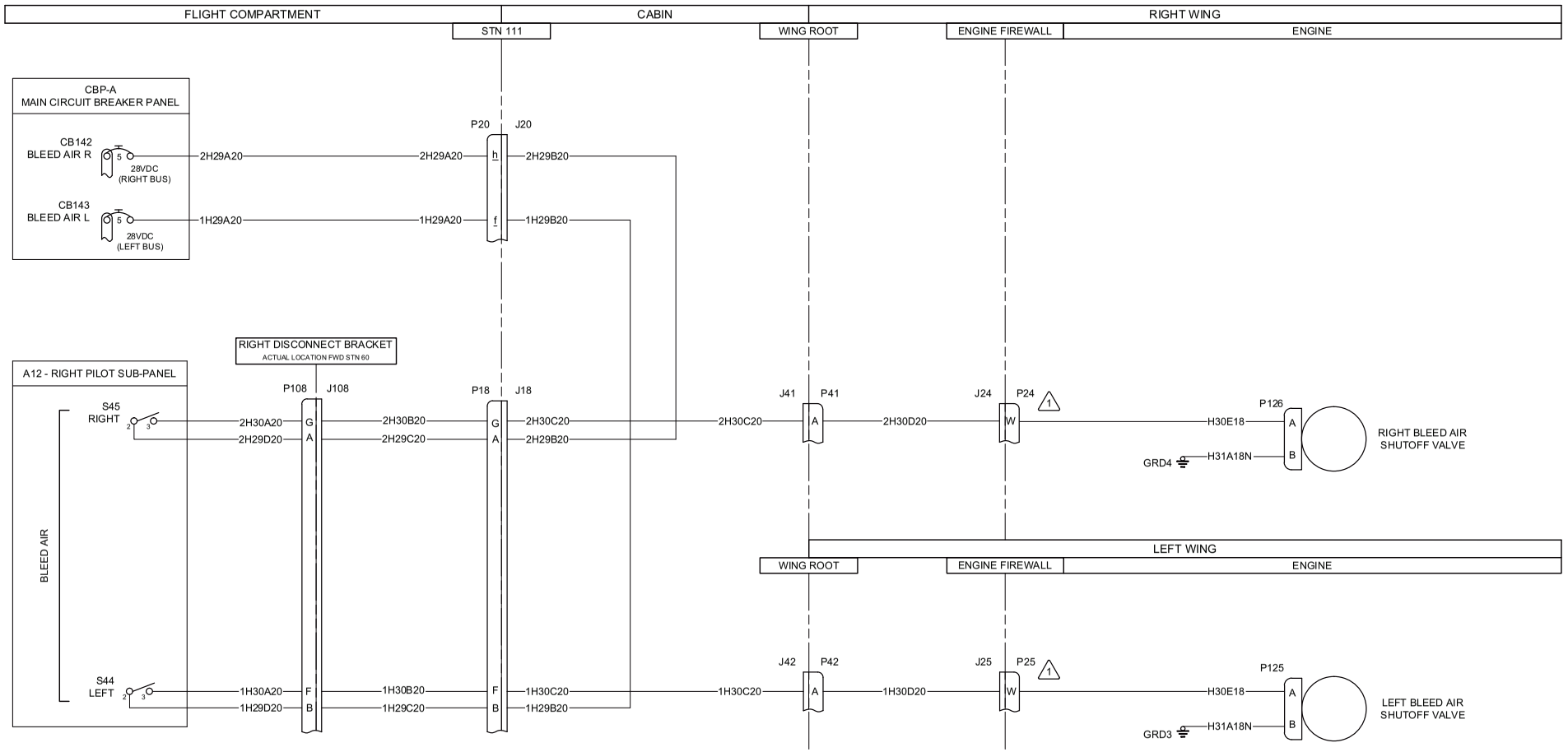


DAU COOLING FAN

SHEET 1 OF 1

C6N1004-35	ISSUE NO.	1

This Page Intentionally Left Blank



NOTES

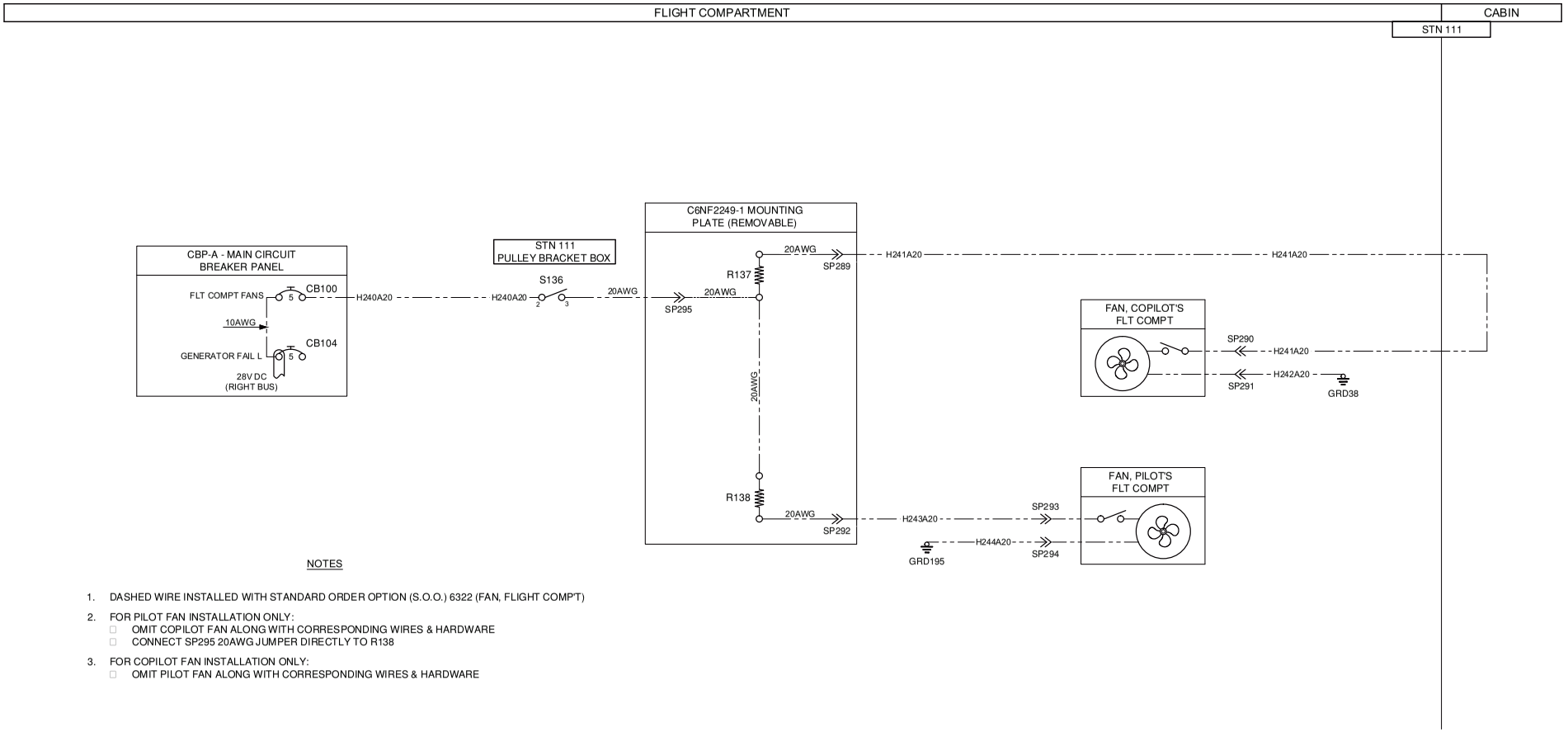
1. HIGH TEMPERATURE RATED WIRE AND CABLE REQUIRED IN ALL AREAS FORWARD THE ENGINE FIREWALL (REFER TO THE GENERAL INFORMATION CHAPTER FOR APPROVED WIRE AND CABLE SPECIFICATIONS)

BLEED AIR SYSTEM

SHEET 1 OF 1

C6N1004-39	ISSUE NO.	1

This Page Intentionally Left Blank



NOTES

1. DASHED WIRE INSTALLED WITH STANDARD ORDER OPTION (S.O.O.) 6322 (FAN, FLIGHT COMP'T)
2. FOR PILOT FAN INSTALLATION ONLY:
   - OMIT COPILOT FAN ALONG WITH CORRESPONDING WIRES & HARDWARE
   - CONNECT SP295 20AWG JUMPER DIRECTLY TO R138
3. FOR COPILOT FAN INSTALLATION ONLY:
   - OMIT PILOT FAN ALONG WITH CORRESPONDING WIRES & HARDWARE

FAN, COOLING & DEMISTING (S.O.O. 6322)									
SHEET 1 OF 1									
C6Z2673	ISSUE NO.	1							

This Page Intentionally Left Blank

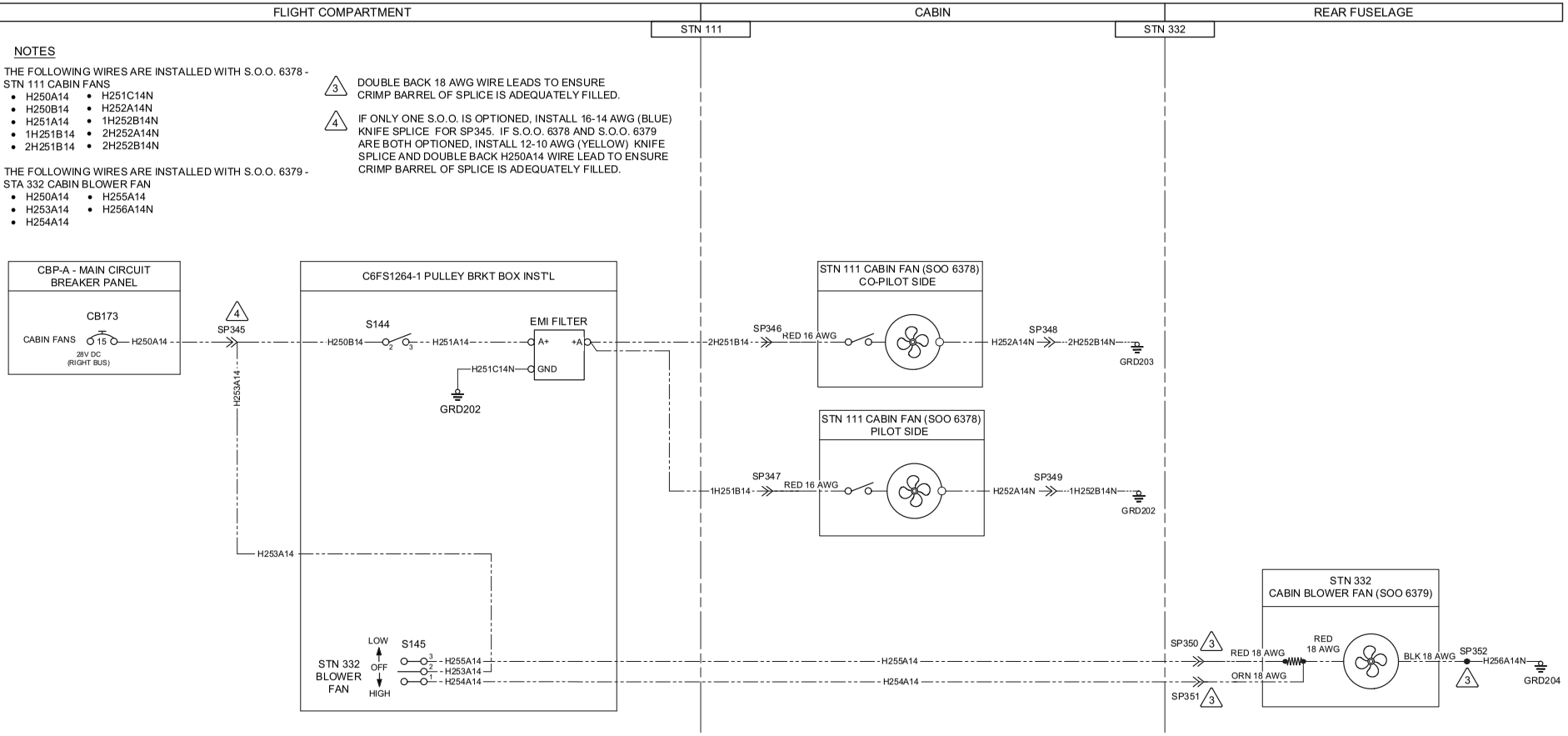
NOTES

1 THE FOLLOWING WIRES ARE INSTALLED WITH S.O.O. 6378 - STN 111 CABIN FANS
- H250A14
- H250B14
- H251A14
- 1H251B14
- 2H251B14
- H251C14N
- H252A14N
- 1H252B14N
- 2H252A14N
- 2H252B14N

2 THE FOLLOWING WIRES ARE INSTALLED WITH S.O.O. 6379 - STA 332 CABIN BLOWER FAN
- H250A14
- H253A14
- H254A14
- H255A14
- H256A14N

△3 DOUBLE BACK 18 AWG WIRE LEADS TO ENSURE CRIMP BARREL OF SPLICE IS ADEQUATELY FILLED.

△4 IF ONLY ONE S.O.O. IS OPTIONED, INSTALL 16-14 AWG (BLUE) KNIFE SPLICE FOR SP345. IF S.O.O. 6378 AND S.O.O. 6379 ARE BOTH OPTIONED, INSTALL 12-10 AWG (YELLOW) KNIFE SPLICE AND DOUBLE BACK H250A14 WIRE LEAD TO ENSURE CRIMP BARREL OF SPLICE IS ADEQUATELY FILLED.

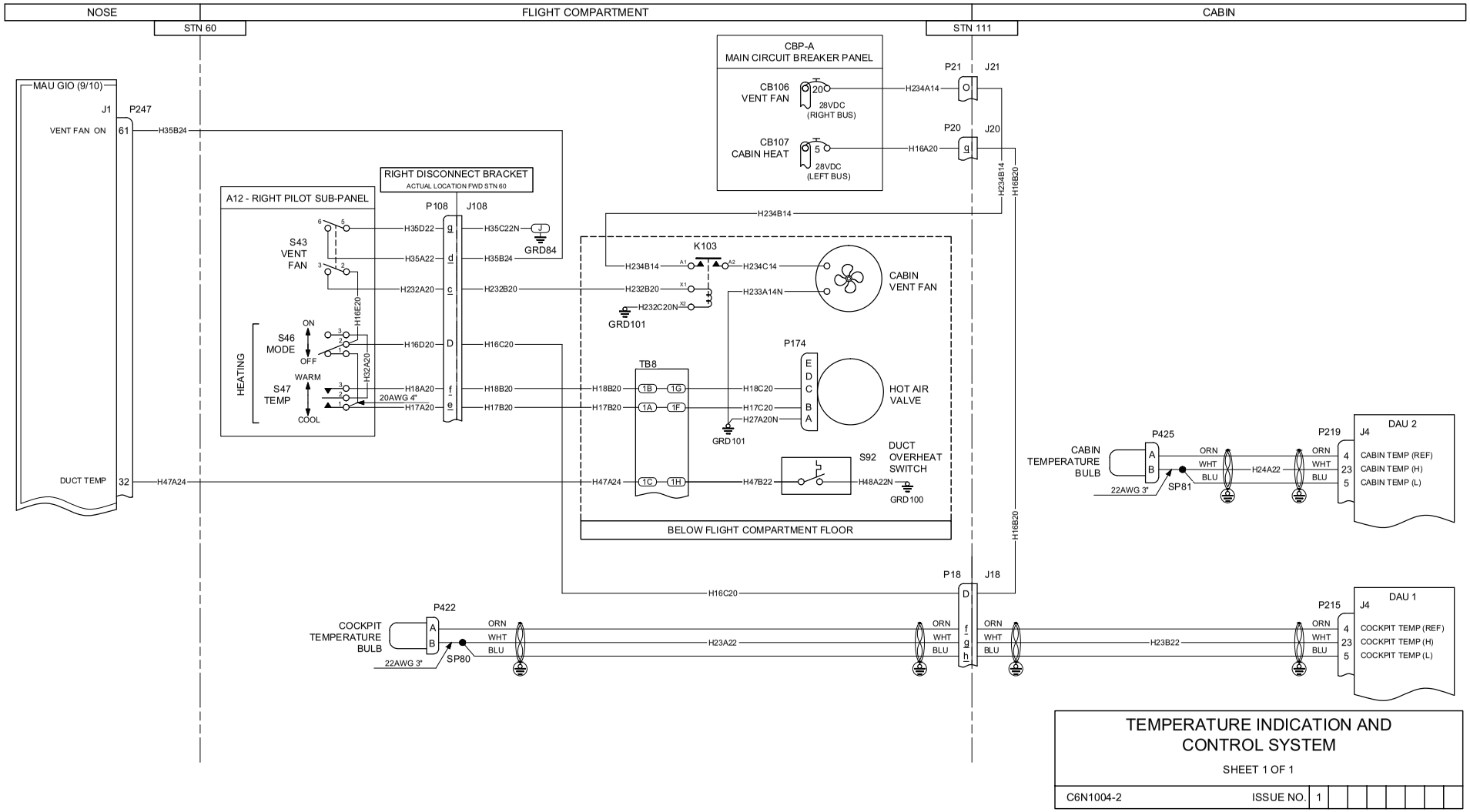


CABIN FANS
(S.O.O. 6378 & S.O.O. 6379)

SHEET 1 OF 1

C6N1004-115A	ISSUE NO.	1	2							

This Page Intentionally Left Blank



TEMPERATURE INDICATION AND CONTROL SYSTEM

SHEET 1 OF 1

C6N1004-2 ISSUE NO. 1

This Page Intentionally Left Blank

CHAPTER 22

AUTO FLIGHT

This Page Intentionally Left Blank

CHAPTER 22 – AUTO FLIGHT

This list shows the effective pages in the manual, with their latest revision date. An asterisk (*) indicates pages added or replaced at the latest revision. Technical changes to the content are indicated by a black vertical line next to the change or the paragraph that contains the change.

REFERENCE	PAGE	DATE
22	1	Sep. 30/14
	2	Blank Page
	* 3	Nov. 30/18
	* 4	Blank Page
	5	Dec. 7/16
	6	Blank Page
22-10-01	1	Dec. 7/16
	2	Dec. 7/16
	3	Dec. 7/16
	4	Dec. 7/16
	5	Dec. 7/16
	6	Dec. 7/16
	7	Dec. 7/16
	8	Dec. 7/16
	9	Dec. 7/16
	10	Dec. 7/16
	11	Dec. 7/16
	12	Dec. 7/16
	13	Dec. 7/16
	14	Blank Page
22-10-02	* 1	Nov. 30/18
	* 2	Nov. 30/18
	* 3	Nov. 30/18
	* 4	Nov. 30/18
	* 5	Nov. 30/18
	* 6	Nov. 30/18
	* 7	Nov. 30/18
	* 8	Nov. 30/18
	* 9	Nov. 30/18
	* 10	Blank Page
22-10-10	1	Dec. 7/16
	2	Blank Page

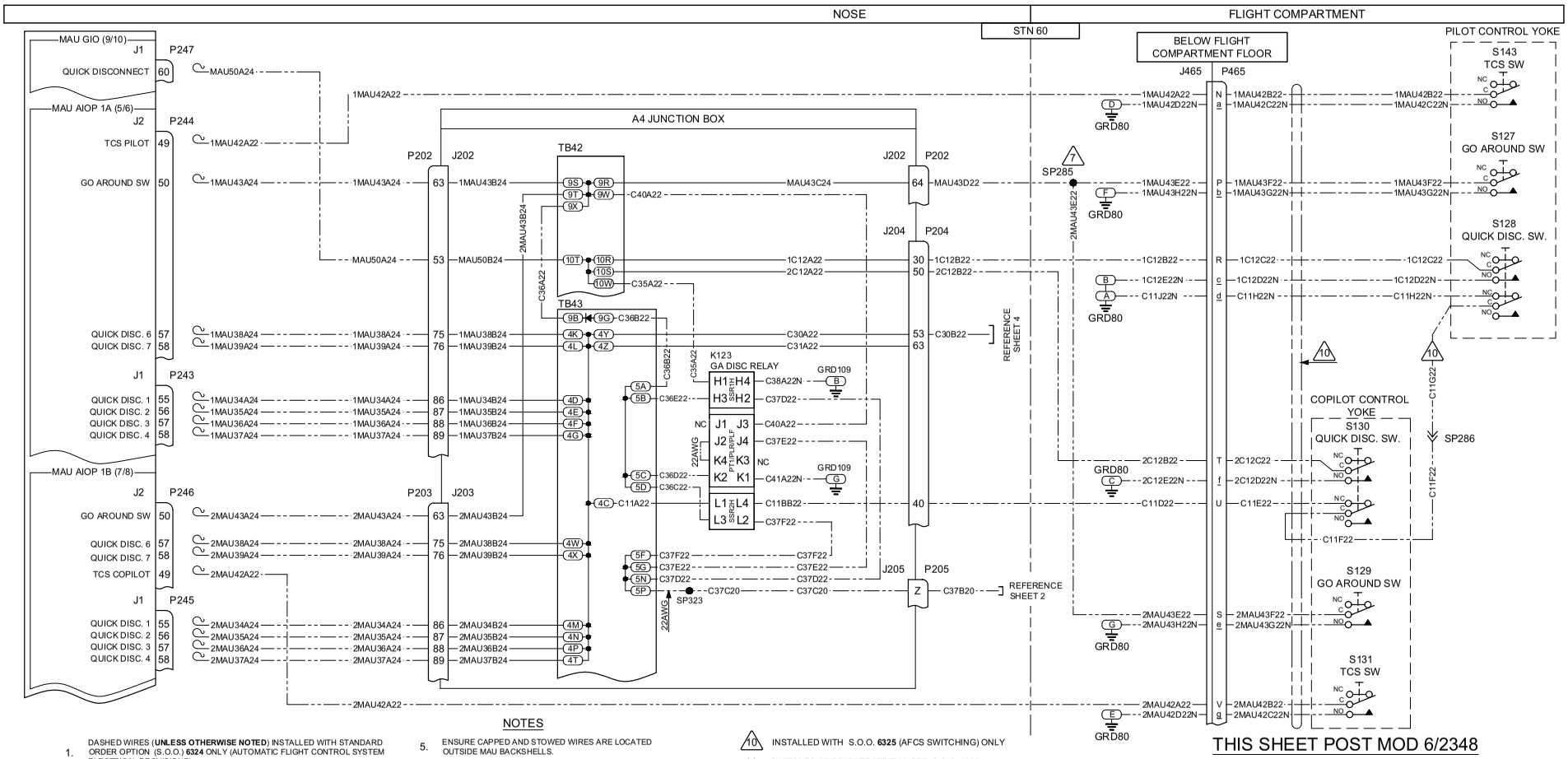
This Page Intentionally Left Blank

CHAPTER 22 – AUTO FLIGHT

TABLE OF CONTENTS

SUBJECT	CHAPTER SECTION SUBJECT
AUTOPILOT	**22-10**
Automatic Flight Control System (AFCS) Electrical Provisions (S.O.O. 6324 & S.O.O. 6325)	22-10-01
Automatic Flight Control System (AFCS) (S.O.O. 6221)	22-10-02
Flight Director – Handwheel Switches (Post Mod 6/2303)	22-10-10

This Page Intentionally Left Blank



THIS SHEET POST MOD 6/2348

NOTES

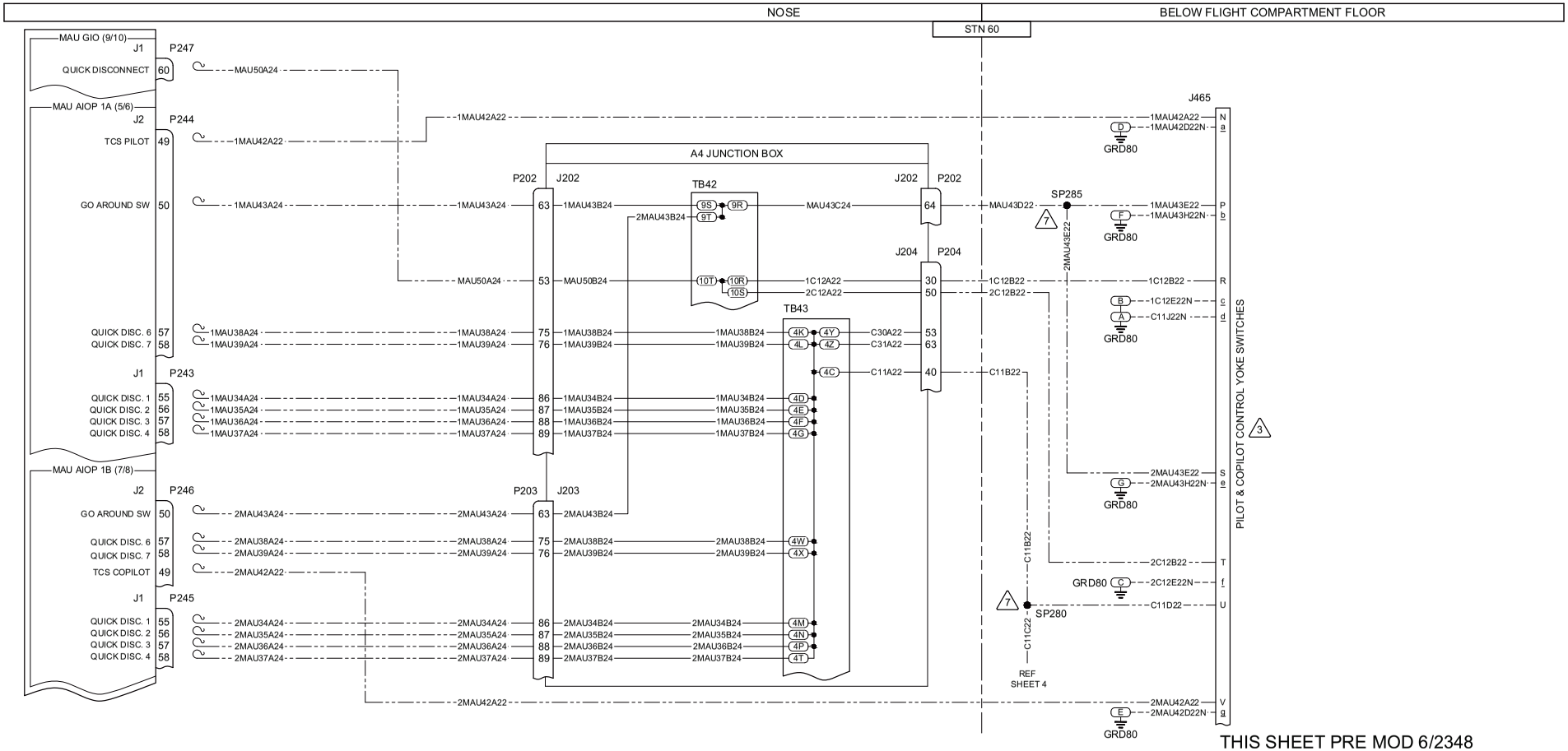
1. DASHED WIRES (**UNLESS OTHERWISE NOTED**) INSTALLED WITH STANDARD ORDER OPTION (S.O.O.) **6324** ONLY (AUTOMATIC FLIGHT CONTROL SYSTEM ELECTRICAL PROVISIONS).
2. AFCS CAN BUS
   - REFER TO GENERAL INFORMATION CHAPTER FOR APPROVED CABLE AND WIRE SPECIFICATIONS
   - REFER TO 20-10-50 FOR AFCS CAN BUS REQUIREMENTS
3. CAP AND STOW CONNECTORS.
4. RESERVED
5. ENSURE CAPPED AND STOWED WIRES ARE LOCATED OUTSIDE MAU BACKSHELLS.
6. ON SIDE OF SPLICE WITH SINGLE WIRE, INSTALL 1 X 14 AWG FILLER WIRE TO ENSURE CRIMP BARREL OF SPLICE IS ADEQUATELY FILLED.
7. ON SIDE OF SPLICE WITH SINGLE WIRE, INSTALL 1 X 18 AWG FILLER WIRE TO ENSURE CRIMP BARREL OF SPLICE IS ADEQUATELY FILLED.
8. IF 22 AWG DRAIN WIRES ARE USED, INSTALL 1 x 18 AWG FILLER WIRE INTO GRD196 CONTACT TO ENSURE CRIMP BARREL OF CONTACT IS ADEQUATELY FILLED.
9. POST MOD 6/2348 ONLY
10. INSTALLED WITH S.O.O. **6325** (AFCS SWITCHING) ONLY
11. SWITCHES ARE INOPERATIVE UNDER S.O.O. **6325**
12. THE CAN BUS STUB CONNECTIONS AT THE ELEVATOR SERVO MUST BE CONSTRUCTED IN IN ACCORDANCE WITH THE TYPE 3 STUB CONSTRUCTION PROCEDURE DESCRIBED IN CHAPTER 20-10-60
13. THE SPLICES AND OVERBRAID SLEEVING AT THE RUDDER AND AILERON SERVO CONNECTIONS MUST BE INSTALLED IN A SIMILAR MANNER AS THE SPLICES AND OVERBRAID SLEEVING USED FOR THE CAN BUS STUB CONNECTIONS AT THE ELEVATOR SERVO. REFERENCE THE TYPE 3 STUB CONSTRUCTION PROCEDURE DESCRIBED IN CHAPTER 20-10-60.

AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS) ELECTRICAL PROVISIONS (S.O.O. 6324 & 6325)

SHEET 1A OF 5

C6N1004-113A ISSUE NO. 1 2 3 4

This Page Intentionally Left Blank



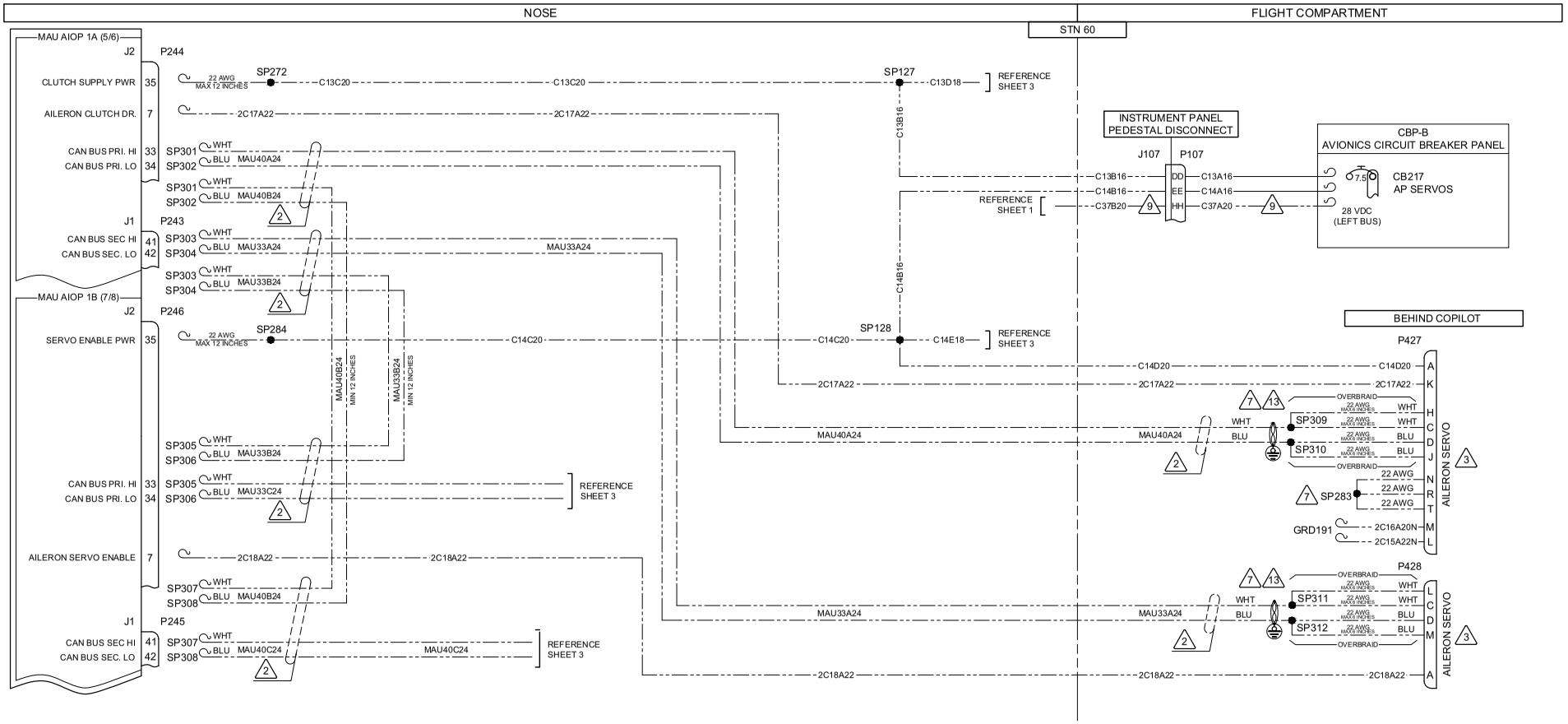
THIS SHEET PRE MOD 6/2348

AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS)
ELECTRICAL PROVISIONS (S.O.O. 6324 & 6325)

SHEET 1B OF 5

C6N1004-113A ISSUE NO. 1 2 3 4

This Page Intentionally Left Blank

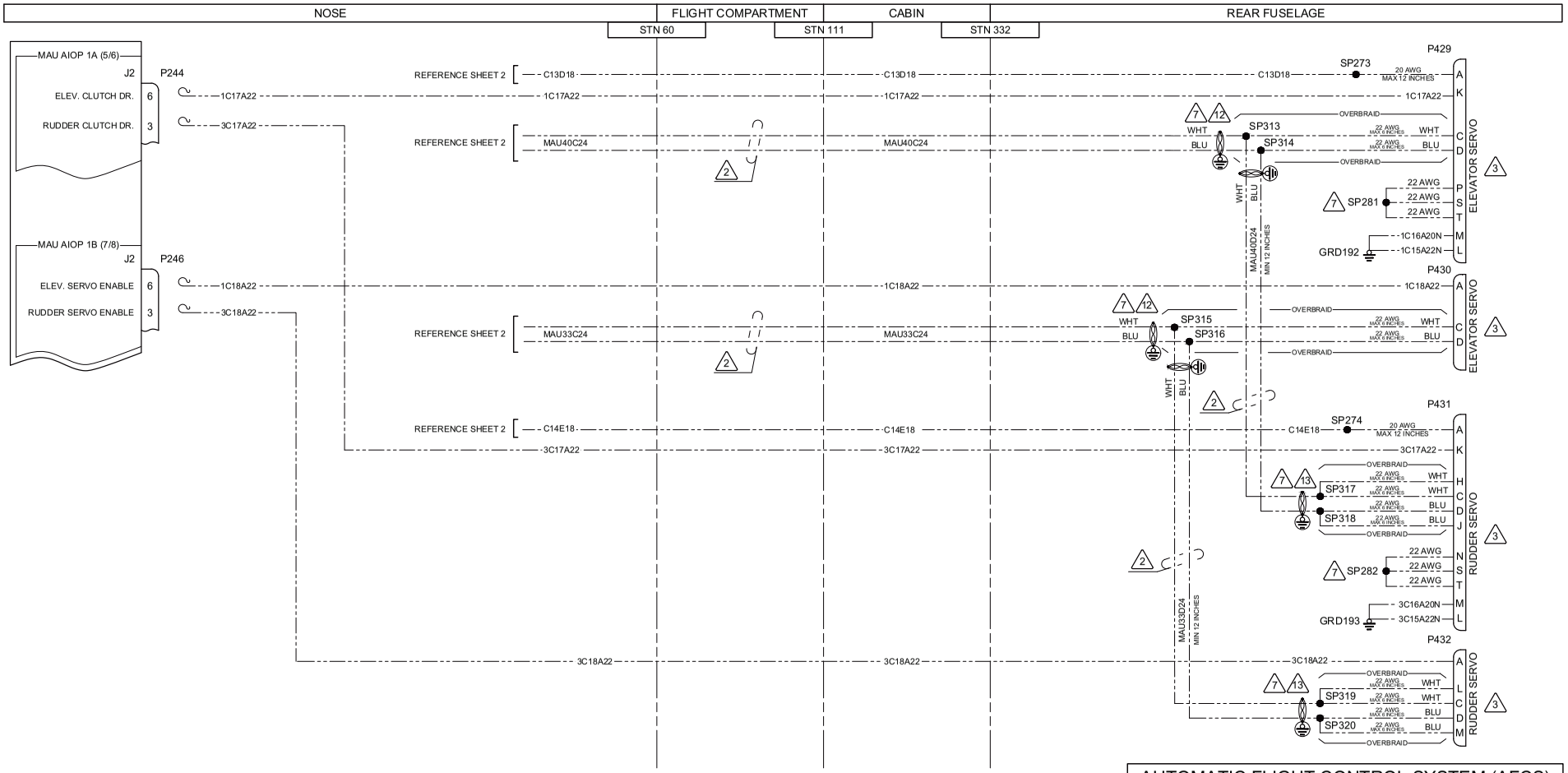


AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS)
ELECTRICAL PROVISIONS (S.O.O. 6324 & 6325)

SHEET 2 OF 5

C6N1004-113A	ISSUE NO.	1	2	3	4				

This Page Intentionally Left Blank

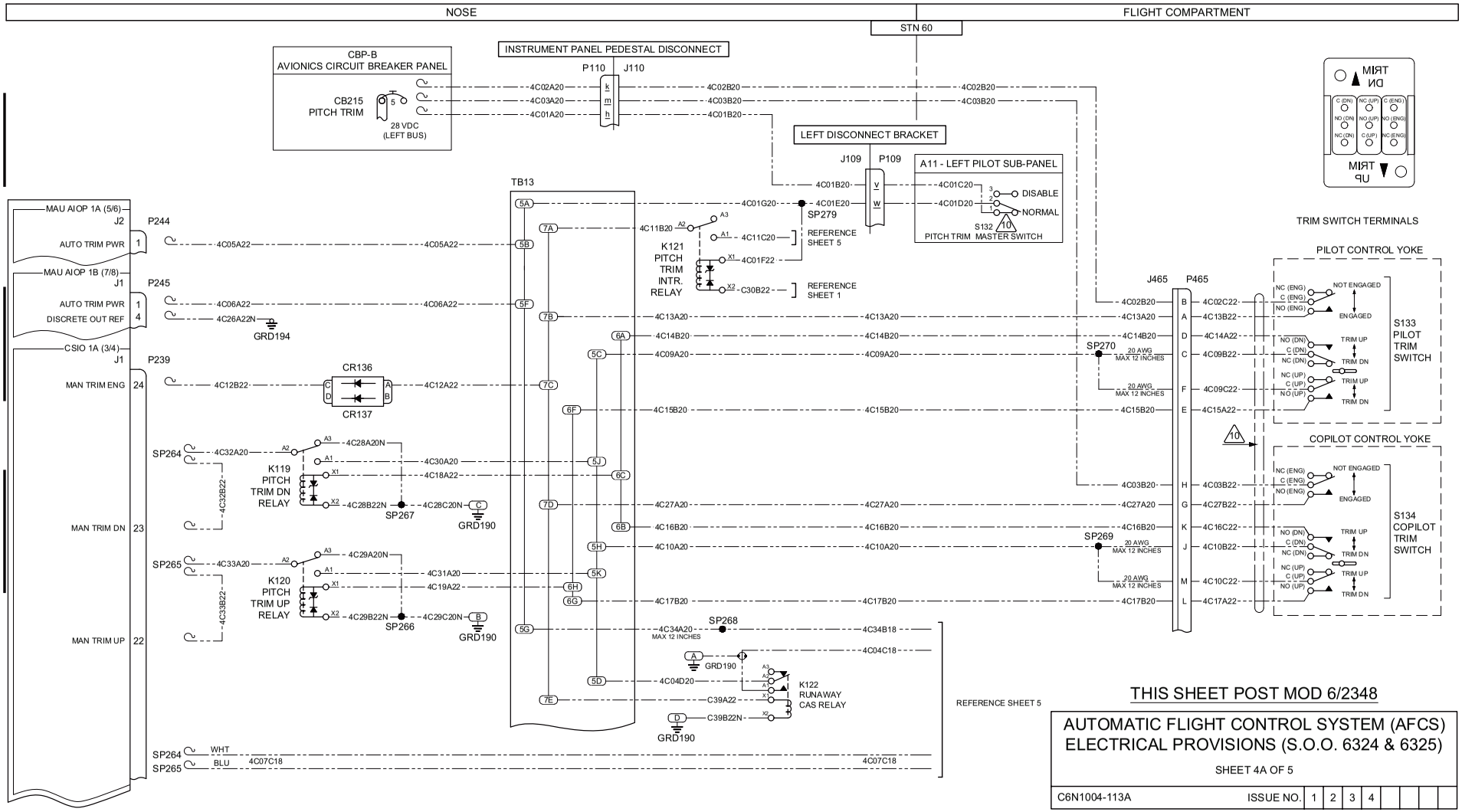


AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS)
ELECTRICAL PROVISIONS (S.O.O. 6324 & 6325)

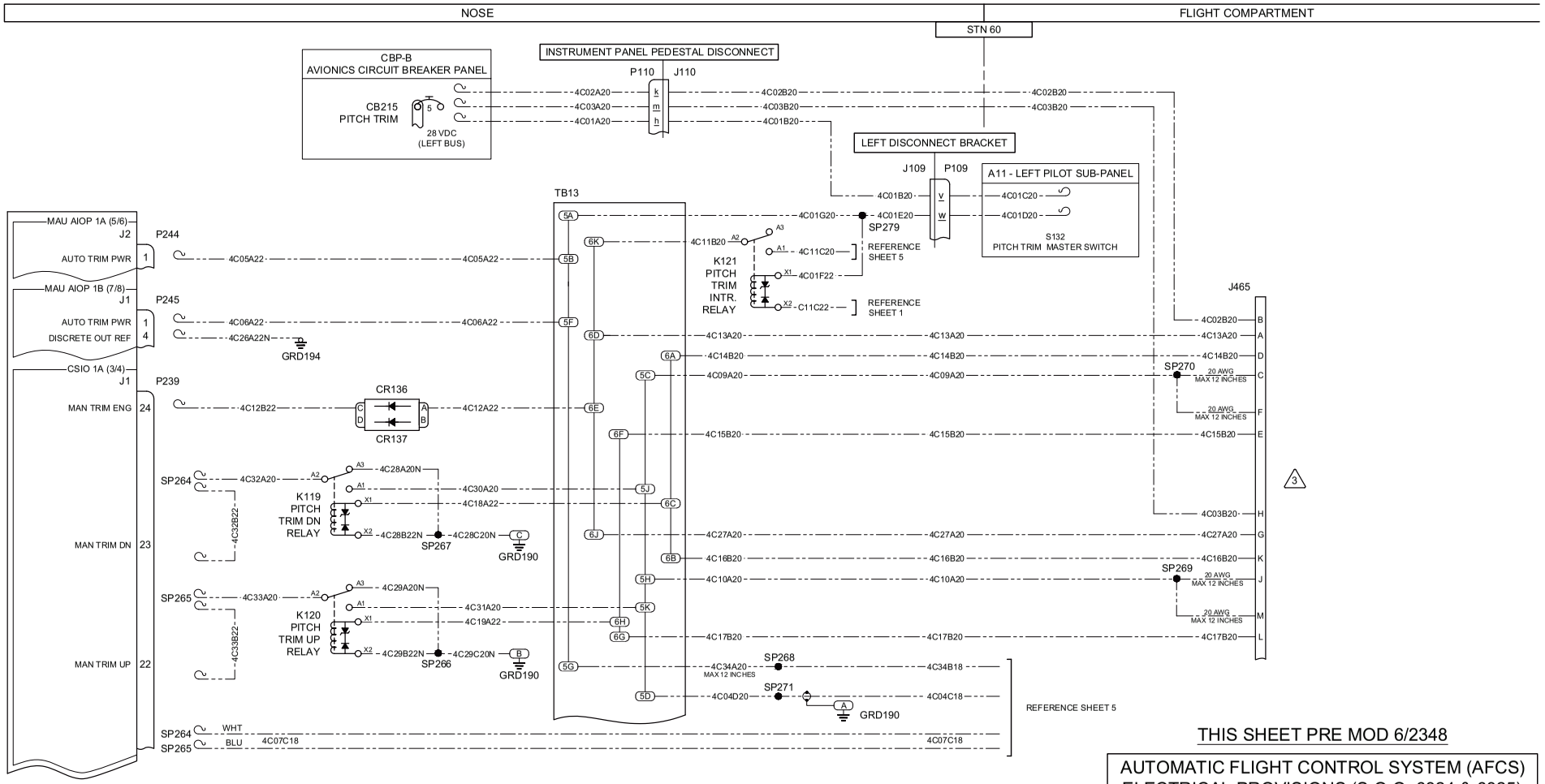
SHEET 3 OF 5

C6N1004-113A ISSUE NO. 1 2 3 4

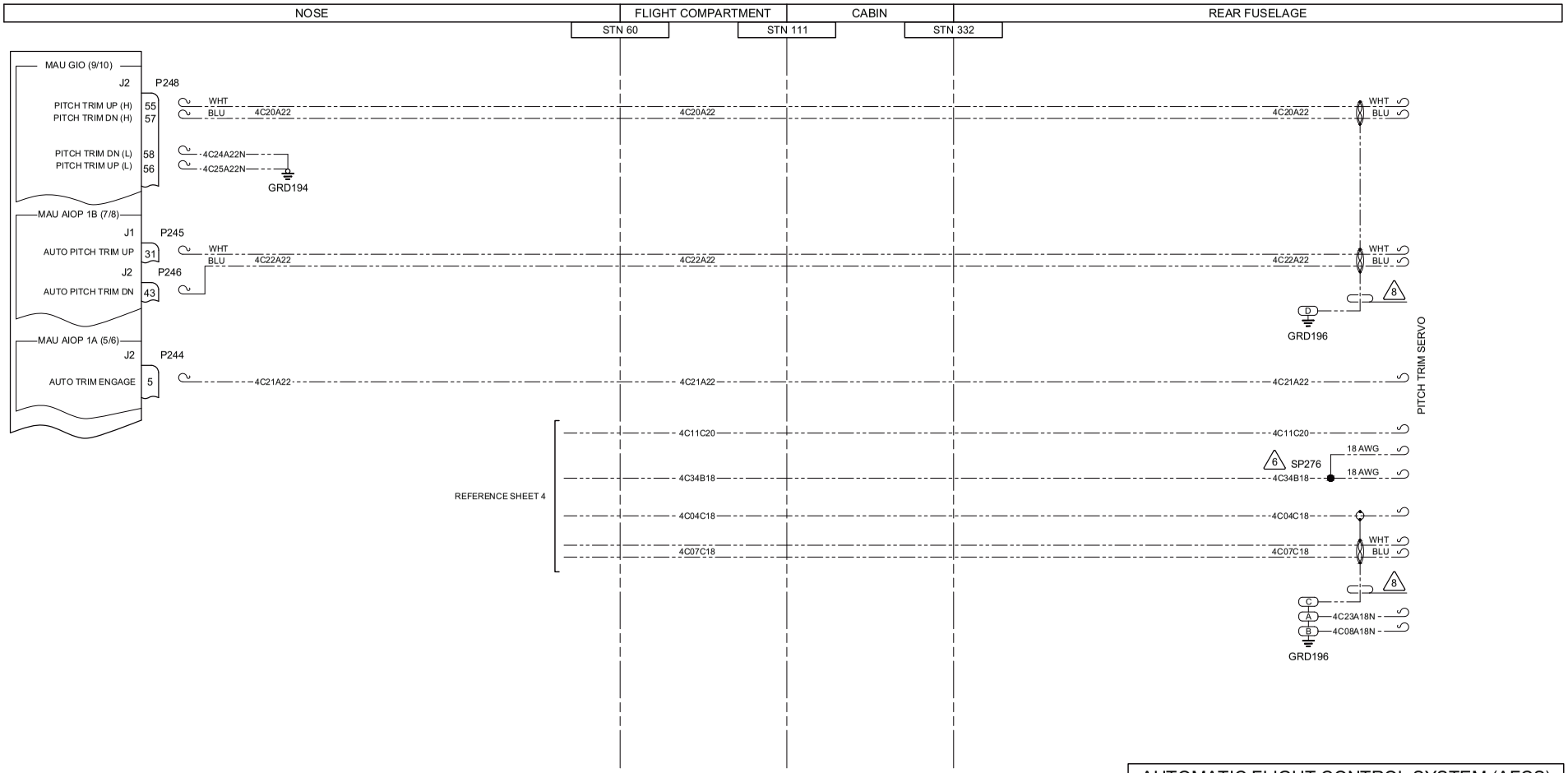
This Page Intentionally Left Blank


NOSE
FLIGHT COMPARTMENT
STN 60
CBP-B
AVIONICS CIRCUIT BREAKER PANEL
CB215
PITCH TRIM
5
28 VDC
(LEFT BUS)
INSTRUMENT PANEL PEDESTAL DISCONNECT
P110
J110
4C02A20
4C03A20
4C01A20
4C02B20
4C03B20
4C01B20
LEFT DISCONNECT BRACKET
J109
P109
A11 - LEFT PILOT SUB-PANEL
4C01C20
4C01D20
DISABLE
NORMAL
S132
10
PITCH TRIM MASTER SWITCH
TRIM SWITCH TERMINALS
TRIM DN
TRIM UP
C (DN)
NC (UP)
C (ENG)
NO (DN)
NO (UP)
NO (ENG)
NC (DN)
C (UP)
NC (ENG)
PILOT CONTROL YOKE
COPILOT CONTROL YOKE
NOT ENGAGED
ENGAGED
TRIM UP
TRIM DN
S133
PILOT
TRIM
SWITCH
S134
COPILOT
TRIM
SWITCH
TB13
J465
P465
4C02B20
4C13A20
4C14B20
4C09A20
4C15B20
4C03B20
4C27A20
4C16B20
4C10A20
4C17B20
4C02C22
4C13B22
4C14A22
4C09B22
4C15A22
4C03B22
4C27B22
4C16C22
4C10B22
4C10C22
4C17A22
SP270
SP269
20 AWG
MAX 12 INCHES
MAU AIOP 1A (5/6)
J2
P244
AUTO TRIM PWR
1
4C05A22
MAU AIOP 1B (7/8)
J1
P245
AUTO TRIM PWR
DISCRETE OUT REF
4
4C06A22
4C26A22N
GRD194
CSIO 1A (3/4)
J1
P239
MAN TRIM ENG
24
4C12B22
4C12A22
CR136
CR137
MAN TRIM DN
23
MAN TRIM UP
22
SP264
SP265
4C32A20
4C33A20
4C32B22
4C33B22
K119
PITCH
TRIM DN
RELAY
K120
PITCH
TRIM UP
RELAY
4C28A20N
4C30A20
4C18A22
4C28B22N
4C28C20N
SP267
GRD190
4C29A20N
4C31A20
4C19A22
4C29B22N
4C29C20N
SP266
SP279
REFERENCE
SHEET 5
K121
PITCH
TRIM
INTR.
RELAY
4C11B20
4C11C20
4C01F22
4C30B22
REFERENCE
SHEET 1
SP268
4C34A20
MAX 12 INCHES
4C34B18
4C04C18
4C04D20
C39A22
C39B22N
K122
RUNAWAY
CAS RELAY
REFERENCE SHEET 5
WHT
BLU
4C07C18
THIS SHEET POST MOD 6/2348
AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS)
ELECTRICAL PROVISIONS (S.O.O. 6324 & 6325)
SHEET 4A OF 5
C6N1004-113A
ISSUE NO. 1 2 3 4

This Page Intentionally Left Blank


NOSE
FLIGHT COMPARTMENT
STN 60
CBP-B
AVIONICS CIRCUIT BREAKER PANEL
CB215
PITCH TRIM
5
28 VDC
(LEFT BUS)
INSTRUMENT PANEL PEDESTAL DISCONNECT
P110
J110
LEFT DISCONNECT BRACKET
J109
P109
A11 - LEFT PILOT SUB-PANEL
S132
PITCH TRIM MASTER SWITCH
SP279
REFERENCE SHEET 5
K121
PITCH TRIM INTR. RELAY
REFERENCE SHEET 1
TB13
J465
MAU AIOP 1A (5/6)
J2
P244
AUTO TRIM PWR
1
MAU AIOP 1B (7/8)
J1
P245
AUTO TRIM PWR
DISCRETE OUT REF
4
GRD194
CSIO 1A (3/4)
J1
P239
MAN TRIM ENG
24
CR136
CR137
SP264
SP265
K119
PITCH TRIM DN RELAY
K120
PITCH TRIM UP RELAY
MAN TRIM DN
23
MAN TRIM UP
22
SP267
SP266
GRD190
SP270
SP269
SP268
SP271
20 AWG
MAX 12 INCHES
WHT
BLU
4C07C18
REFERENCE SHEET 5
THIS SHEET PRE MOD 6/2348
AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS)
ELECTRICAL PROVISIONS (S.O.O. 6324 & 6325)
SHEET 4B OF 5
C6N1004-113A
ISSUE NO. 1 2 3 4

This Page Intentionally Left Blank

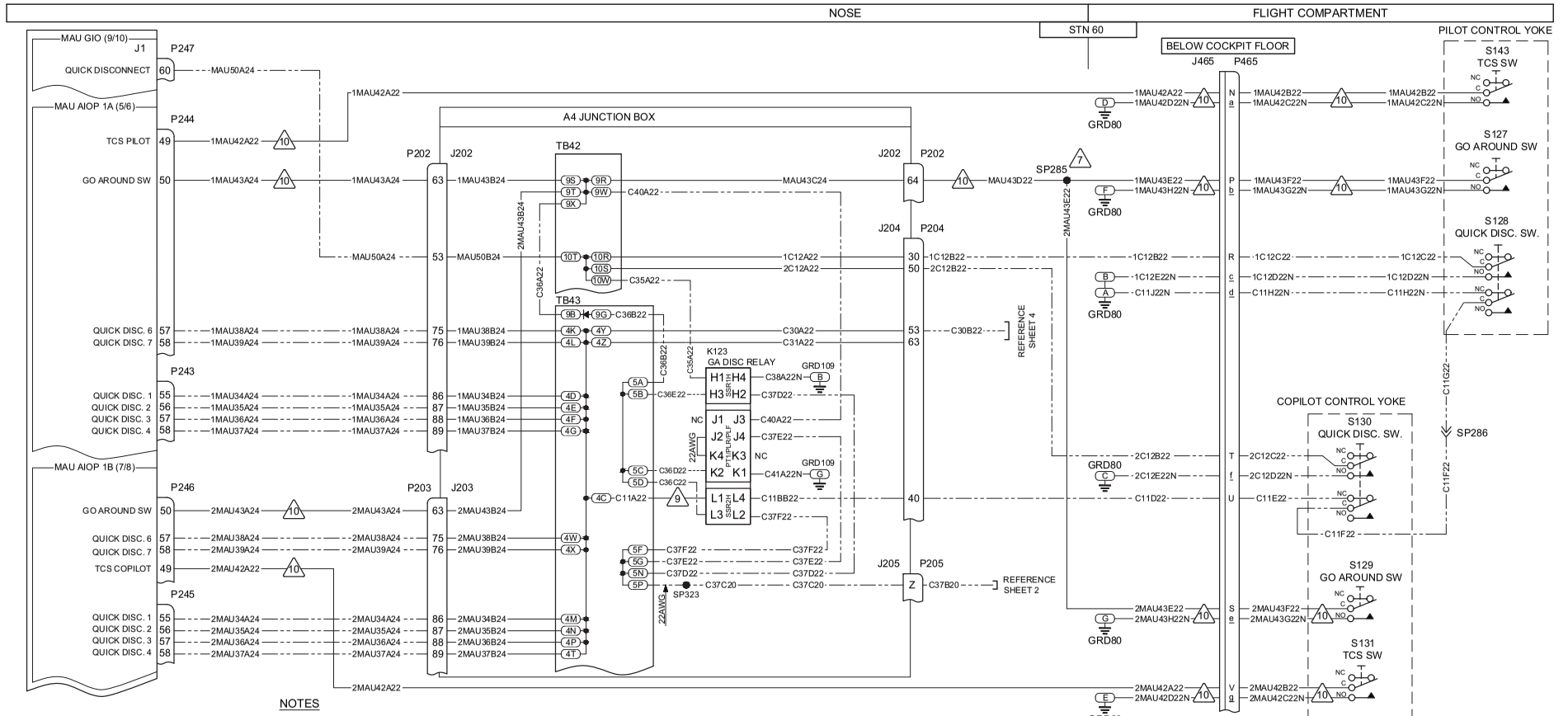


AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS)
ELECTRICAL PROVISIONS (S.O.O. 6324 & 6325)

SHEET 5 OF 5

C6N1004-113A	ISSUE NO.	1	2	3	4				

This Page Intentionally Left Blank



NOTES

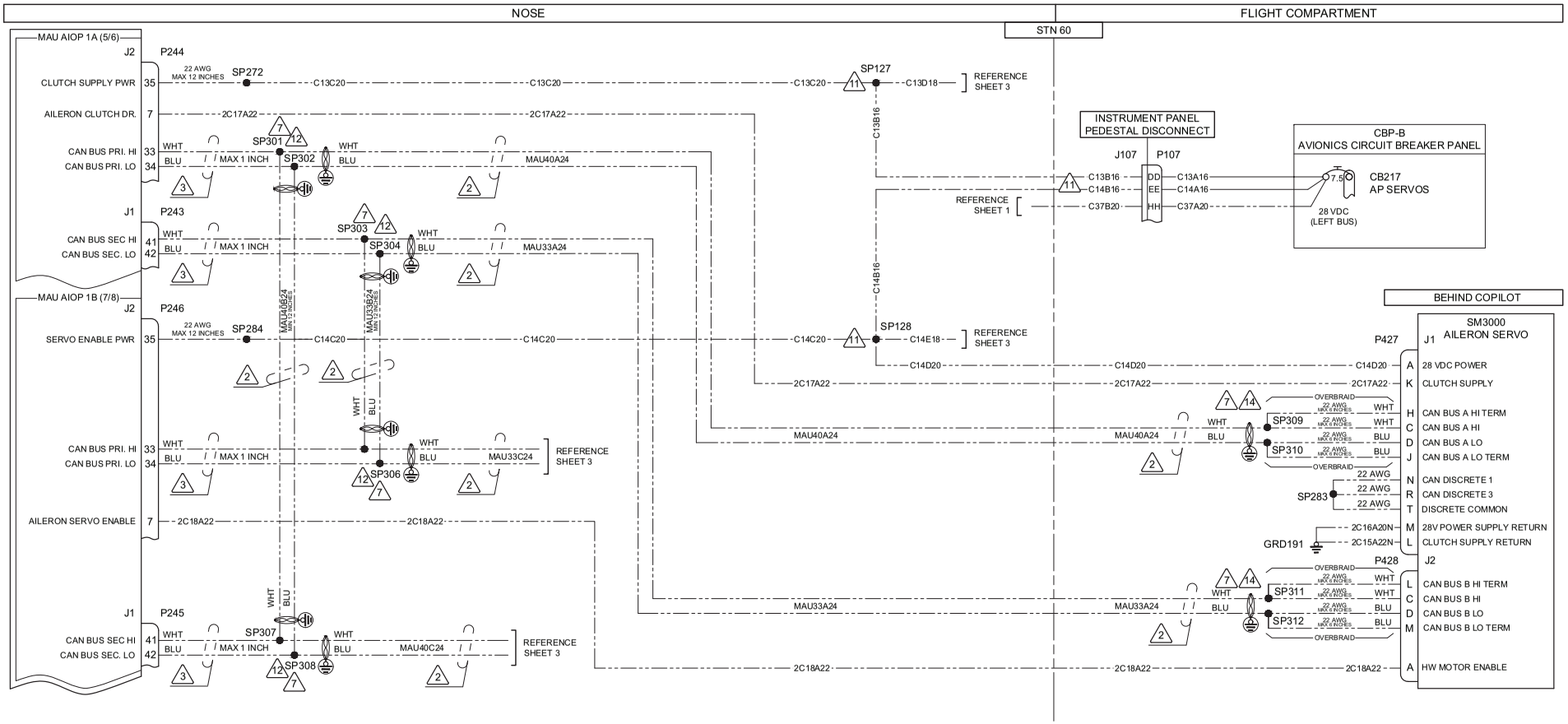
1. DASHED WIRES INSTALLED WITH STANDARD ORDER OPTION (S.O.O) 6221 ONLY (AUTOMATIC FLIGHT CONTROL SYSTEM)
2. AFCS CAN BUS
   - REFER TO GENERAL INFORMATION CHAPTER FOR APPROVED WIRE AND CABLE SPECIFICATIONS
   - REFER TO 20-10-50 FOR AFCS CAN BUS REQUIREMENTS
3. THINWALL WIRE (REFER TO GENERAL INFORMATION CHAPTER FOR APPROVED WIRE AND CABLE SPECIFICATIONS)
4. RESERVED
5. - CRIMP ONE 18 AWG WIRE AND ONE 22 AWG WIRE INTO ONE END OF SPLICE - CRIMP ONE 20 AWG WIRE INTO OPPOSITE END OF SPLICE
6. ON SIDE OF SPLICE WITH SINGLE WIRE, INSTALL 1 X 14 AWG FILLER WIRE TO ENSURE CRIMP BARREL OF SPLICE IS ADEQUATELY FILLED
7. ON SIDE OF SPLICE WITH SINGLE WIRE, INSTALL 1 X 18 AWG FILLER WIRE TO ENSURE CRIMP BARREL OF SPLICE IS ADEQUATELY FILLED.
8. IF 22 AWG DRAIN WIRES ARE USED, INSTALL 1 x 18 AWG FILLER WIRE INTO GRD196 CONTACT TO ENSURE CRIMP BARREL OF CONTACT IS ADEQUATELY FILLED.
9. WIRE CONNECTED DIRECTLY TO J204-40 WHEN S.O.O. 6221 IS NOT INSTALLED
10. FLIGHT DIRECTOR WIRES (MOD 6/2303)
11. REPLACES STANDARD (POST MOD 6/2303) BUILD FLIGHT DIRECTOR WIRES
12. THE CAN BUS STUB CONNECTIONS AT THE AIOP MUST BE CONSTRUCTED IN ACCORDANCE WITH THE TYPE 2 STUB CONSTRUCTION PROCEDURE DESCRIBED IN CHAPTER 20-10-60.
13. THE CAN BUS STUB CONNECTIONS AT THE ELEVATOR SERVO MUST BE CONSTRUCTED IN ACCORDANCE WITH THE TYPE 3 STUB CONSTRUCTION PROCEDURE DESCRIBED IN CHAPTER 20-10-60.
14. THE SPLICES AND OVERBRAID SLEEVING AT THE RUDDER AND AILERON SERVO CONNECTIONS MUST BE INSTALLED IN A SIMILAR MANNER AS THE SPLICES AND OVERBRAID SLEEVING USED FOR THE CAN BUS STUB CONNECTIONS AT THE ELEVATOR SERVO. REFERENCE THE TYPE 2 STUB CONSTRUCTION PROCEDURE DESCRIBED IN CHAPTER 20-10-60.

AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS) (S.O.O. 6221)

SHEET 1 OF 5

C6N1004-112C ISSUE NO. 1 2 3

This Page Intentionally Left Blank

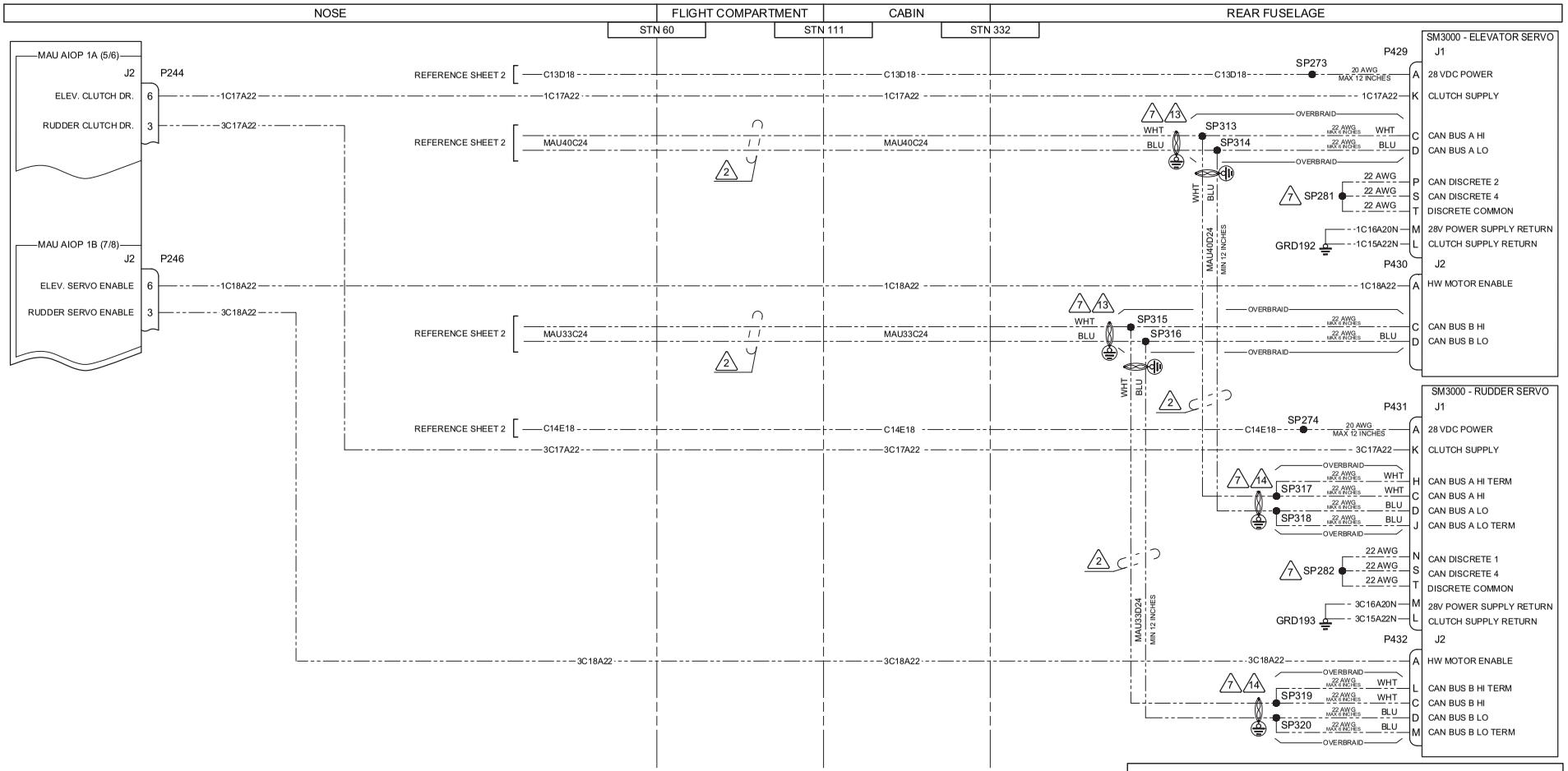


AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS) (S.O.O. 6221)

SHEET 2 OF 5

C6N1004-112C	ISSUE NO.	1	2	3					

This Page Intentionally Left Blank

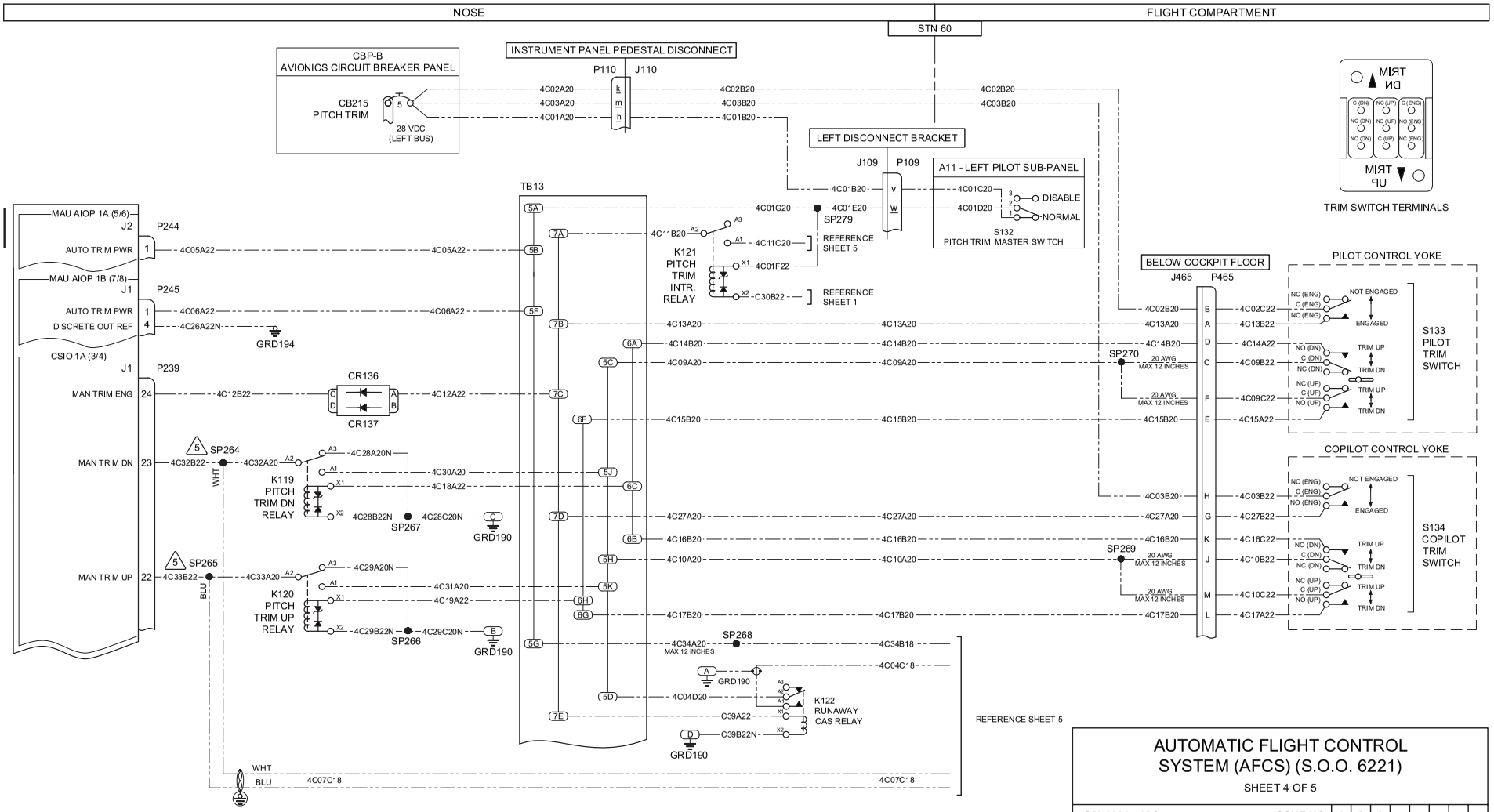


AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS) (S.O.O. 6221)

SHEET 3 OF 5

C6N1004-112C ISSUE NO. 1 2 3

This Page Intentionally Left Blank

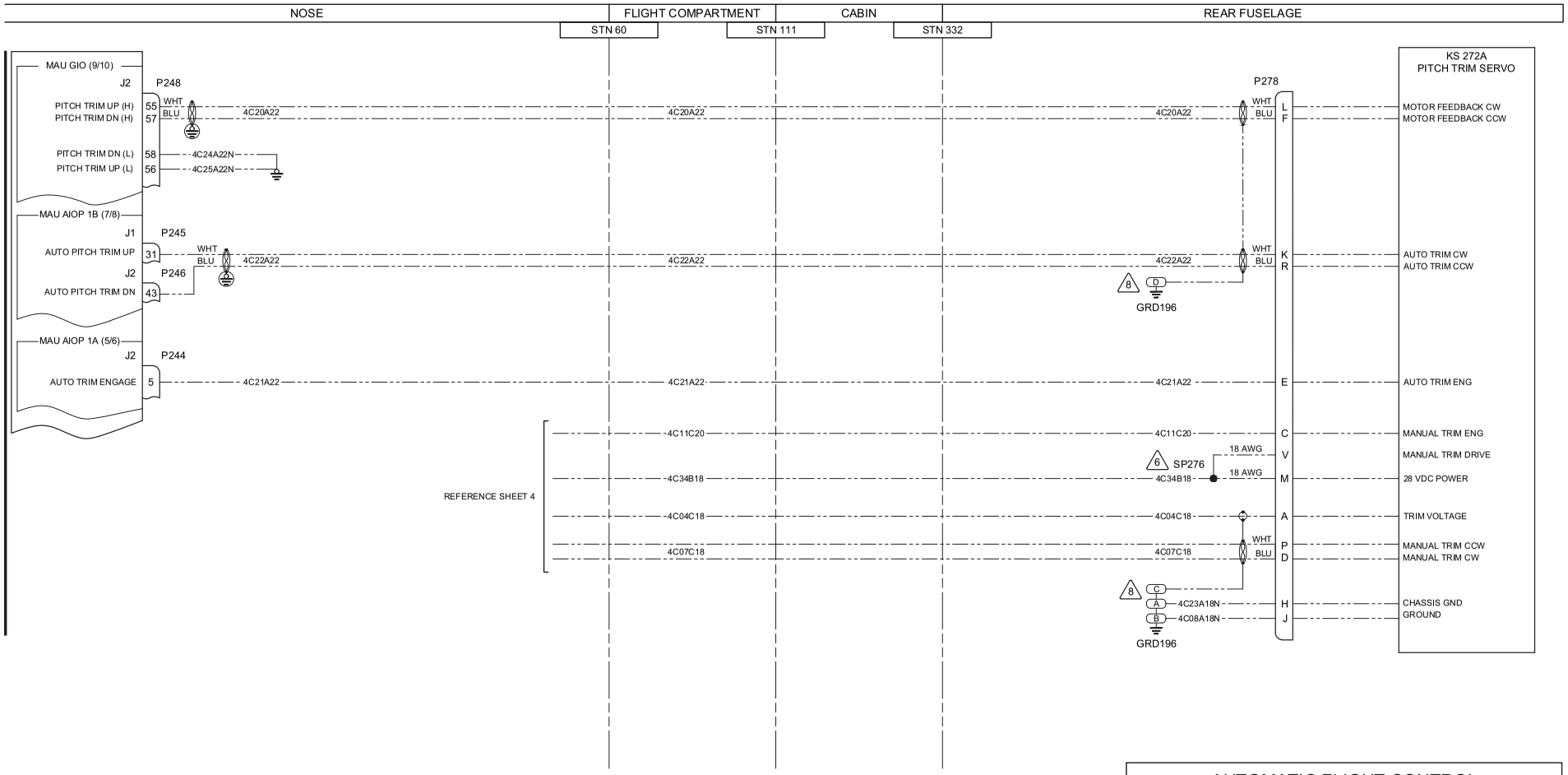


AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS) (S.O.O. 6221)

SHEET 4 OF 5

C6N1004-112C ISSUE NO. 1 2 3

This Page Intentionally Left Blank

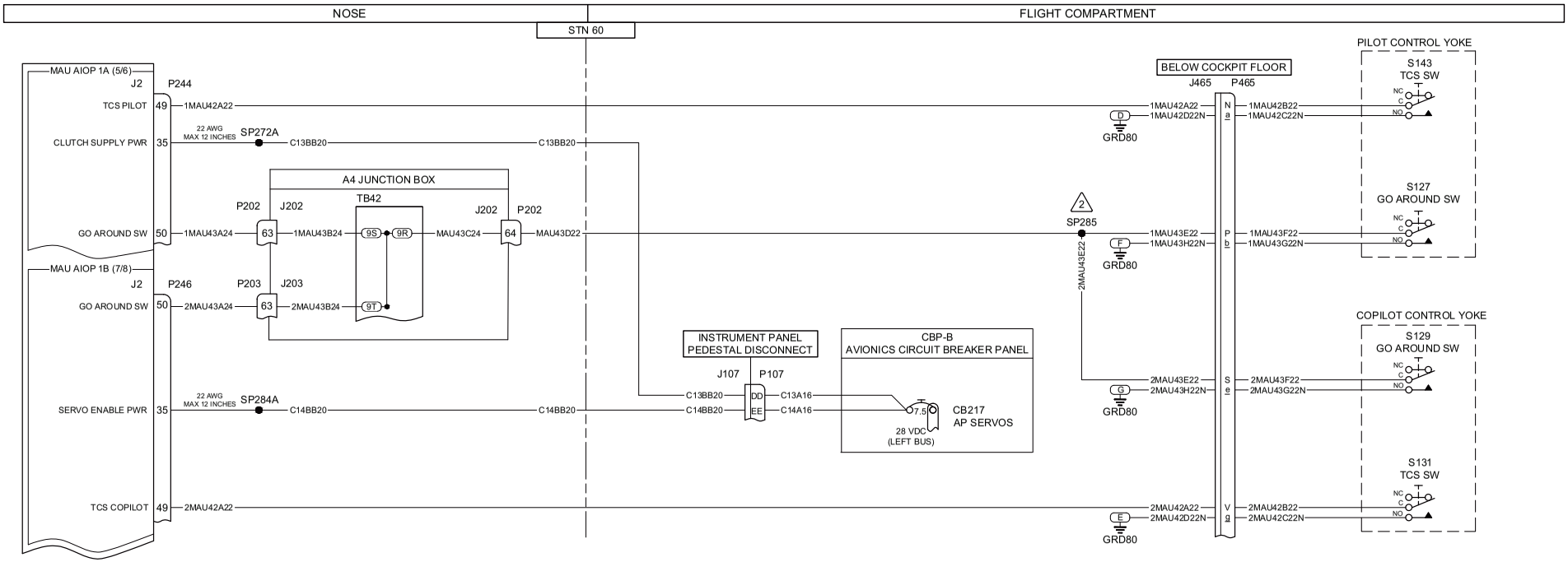


AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS) (S.O.O. 6221)

SHEET 5 OF 5

C6N1004-112C ISSUE NO. 1 2 3

This Page Intentionally Left Blank



NOTES

1. WIRING ONLY FOLLOWED FOR **POST MOD 6/2303**
   REFER TO 22-10-02 WHEN STANDARD ORDER OPTION
   (S.O.O.) 6221 (SERIES 400 AFCS (AUTOPILOT)) IS INSTALLED

2 ON SIDE OF SPLICE WITH SINGLE WIRE, INSTALL 1 X 18 AWG FILLER WIRE TO ENSURE CRIMP BARREL OF SPLICE IS ADEQUATELY FILLED.

FLIGHT DIRECTOR - HANDWHEEL SWITCHES (POST MOD 6/2303)			
SHEET 1 OF 1			
C6N1004-114NC	ISSUE NO.	1	

This Page Intentionally Left Blank

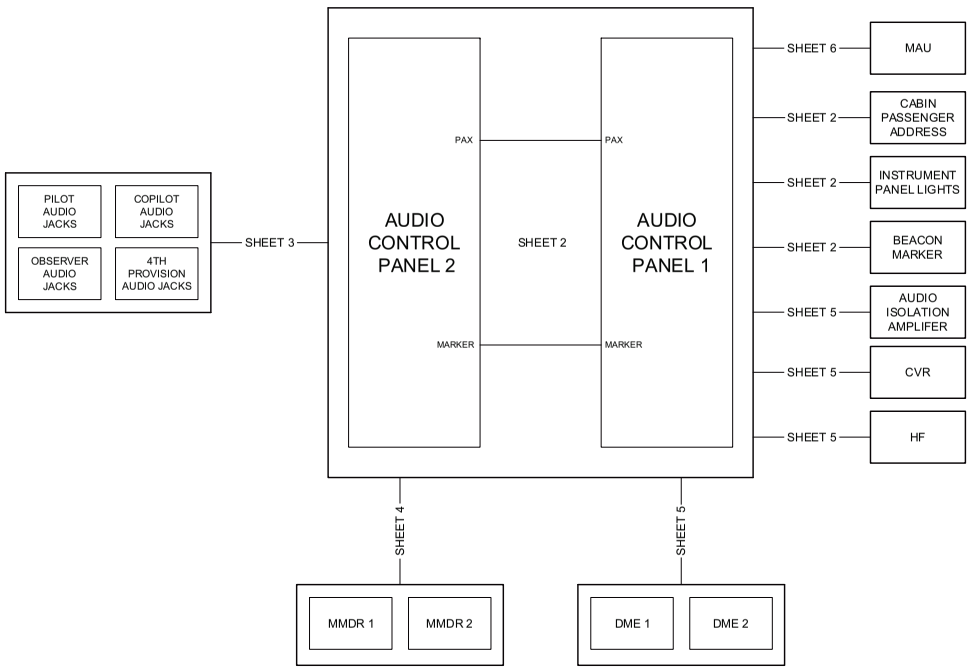
CHAPTER 23 – COMMUNICATIONS

This list shows the effective pages in the manual, with their latest revision date. An asterisk (*) indicates pages added or replaced at the latest revision. Technical changes to the content are indicated by a black vertical line next to the change or the paragraph that contains the change.

REFERENCE	PAGE	DATE	REFERENCE	PAGE	DATE
23	1	Sep. 30/14	23-53-00	* 1	Dec. 7/16
	2	Blank Page		* 2	Dec. 7/16
	* 3	Dec. 7/16		* 3	Dec. 7/16
	* 4	Blank Page		* 4	Dec. 7/16
	5	Sep. 30/14		* 5	Dec. 7/16
	6	Blank Page		* 6	Dec. 7/16
23-11-00	1	Sep. 30/14		* 7	Dec. 7/16
	2	Sep. 30/14		* 8	Dec. 7/16
	3	Sep. 30/14		* 9	Dec. 7/16
	4	Blank Page		* 10	Dec. 7/16
23-13-00	1	Sep. 30/14		* 11	Dec. 7/16
	2	Blank Page		* 12	Blank Page
23-16-00	1	Sep. 30/14	23-54-00	1	Sep. 30/14
	2	Blank Page		2	Blank Page

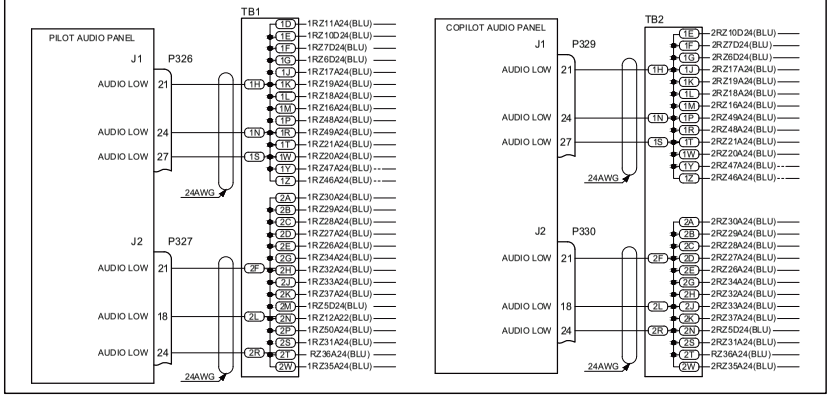
This Page Intentionally Left Blank

BLOCK DIAGRAM - AUDIO CONTROL PANEL INTERCONNECTS



NOTES

1. ENSURE AUDIO JACKS ARE ISOLATED FROM AIRCRAFT GROUND BY USING AN ISOLATED WASHER (REFER TO EQUIPMENT LIST FOR PART NUMBERS)
2. CONNECTING TEL MODE PIN (J2-7) TO PROGRAM GROUND PIN (J2-30) ENABLES FULL DUPLEX MODE ON COM 5 POSITION, **JUMPERS ONLY CONNECTED WHEN (S.O.O.) 6298 (SATELLITE COMMUNICATION SYSTEM) IS INCORPORATED**
3. TB1/TB2 CONNECTIONS, THE UNSHIELDED PORTION OF ALL WIRES CONNECTED TO TB1 AND TB2 IS RESTRICTED TO A MAXIMUM LENGTH OF 6"



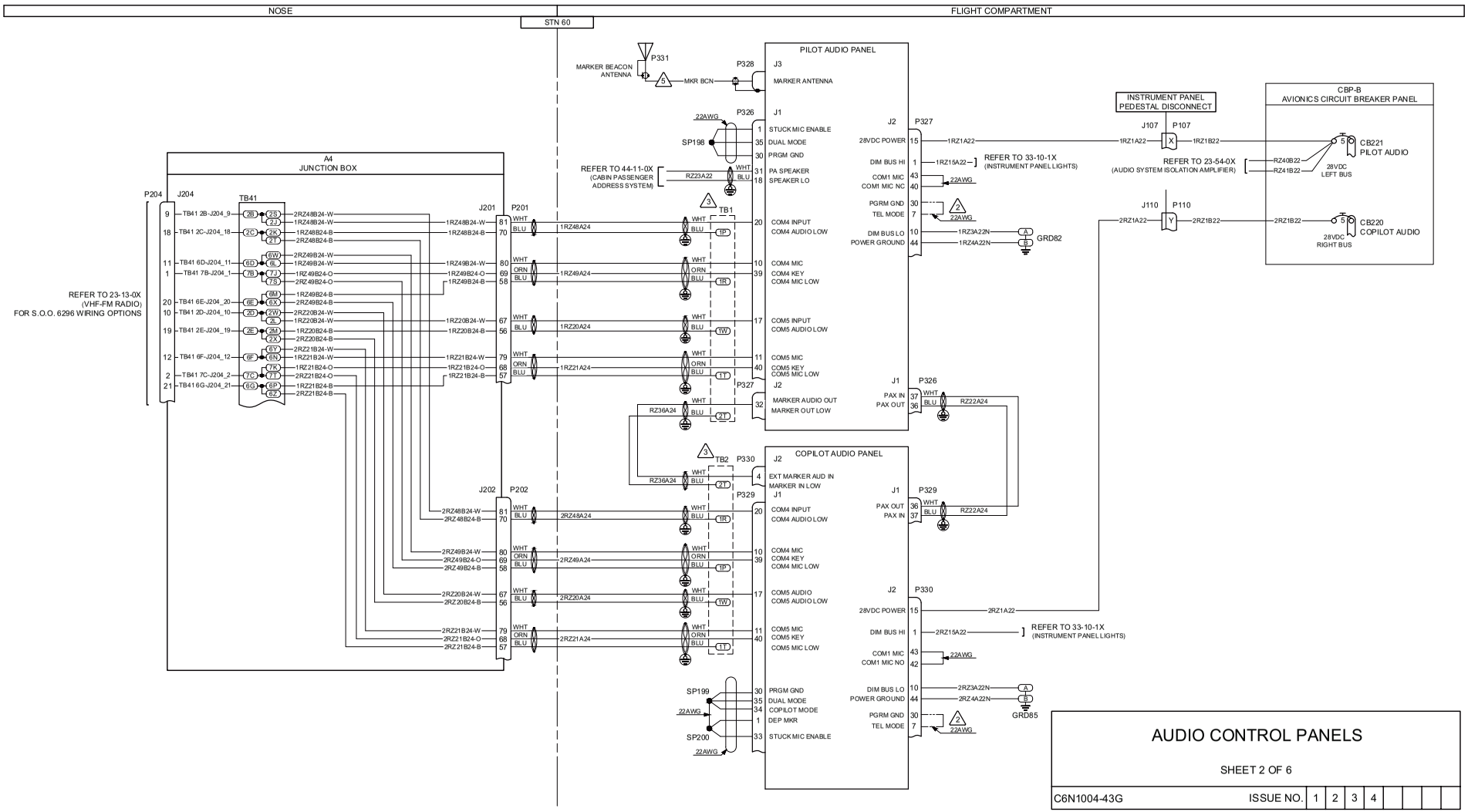
4. DASHED WIRE INSTALLED WITH STANDARD ORDER OPTION (S.O.O.) 6215 (HF COMMUNICATION SYSTEM) OR (S.O.O.) 6296 (VHF-FM RADIO)
5. COAXIAL CABLE - (REFER TO THE GENERAL INFORMATION CHAPTER FOR APPROVED CABLE SPECIFICATIONS AND LENGTH RESTRICTIONS)
6. PRE MOD 6/2114: WIRE SHIELDS RZ44A24, RZ45A24 NOT GROUNDED TO BACKSHELL
   POST MOD 6/2114: WIRE SHIELDS RZ44A24, RZ45A24 GROUNDED TO BACKSHELL
7. WIRES INSTALLED WITH STANDARD ORDER OPTION (S.O.O.) 6229 ONLY (FOURTH CREW INTERCOM)
8. WIRES CAPPED AND STOWED WHEN (S.O.O.) 6292 (REMOVAL OF CVR) IS INCORPORATED

AUDIO CONTROL PANELS

SHEET 1 OF 6

C6N1004-43G	ISSUE NO.	1	2	3	4				

This Page Intentionally Left Blank

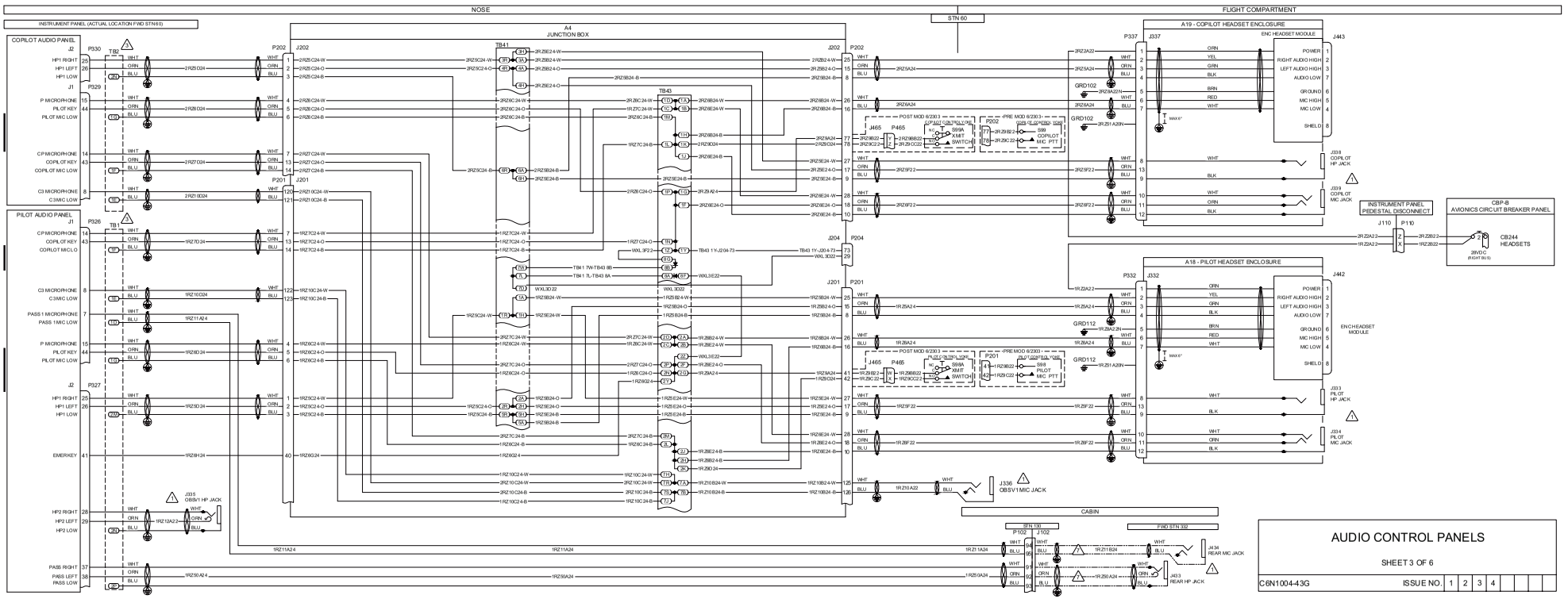


AUDIO CONTROL PANELS

SHEET 2 OF 6

C6N1004-43G ISSUE NO. 1 2 3 4

This Page Intentionally Left Blank

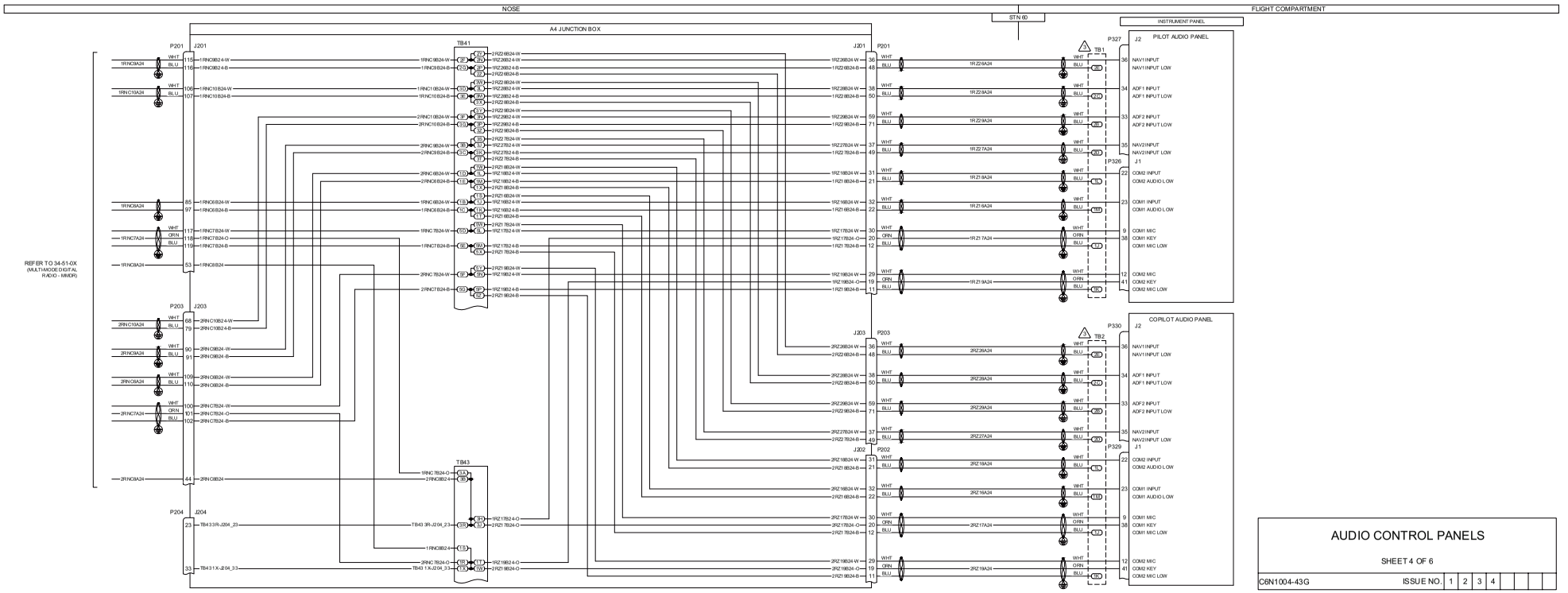


AUDIO CONTROL PANELS

SHEET 3 OF 6

C6N1004-43G ISSUE NO. 1 2 3 4

This Page Intentionally Left Blank

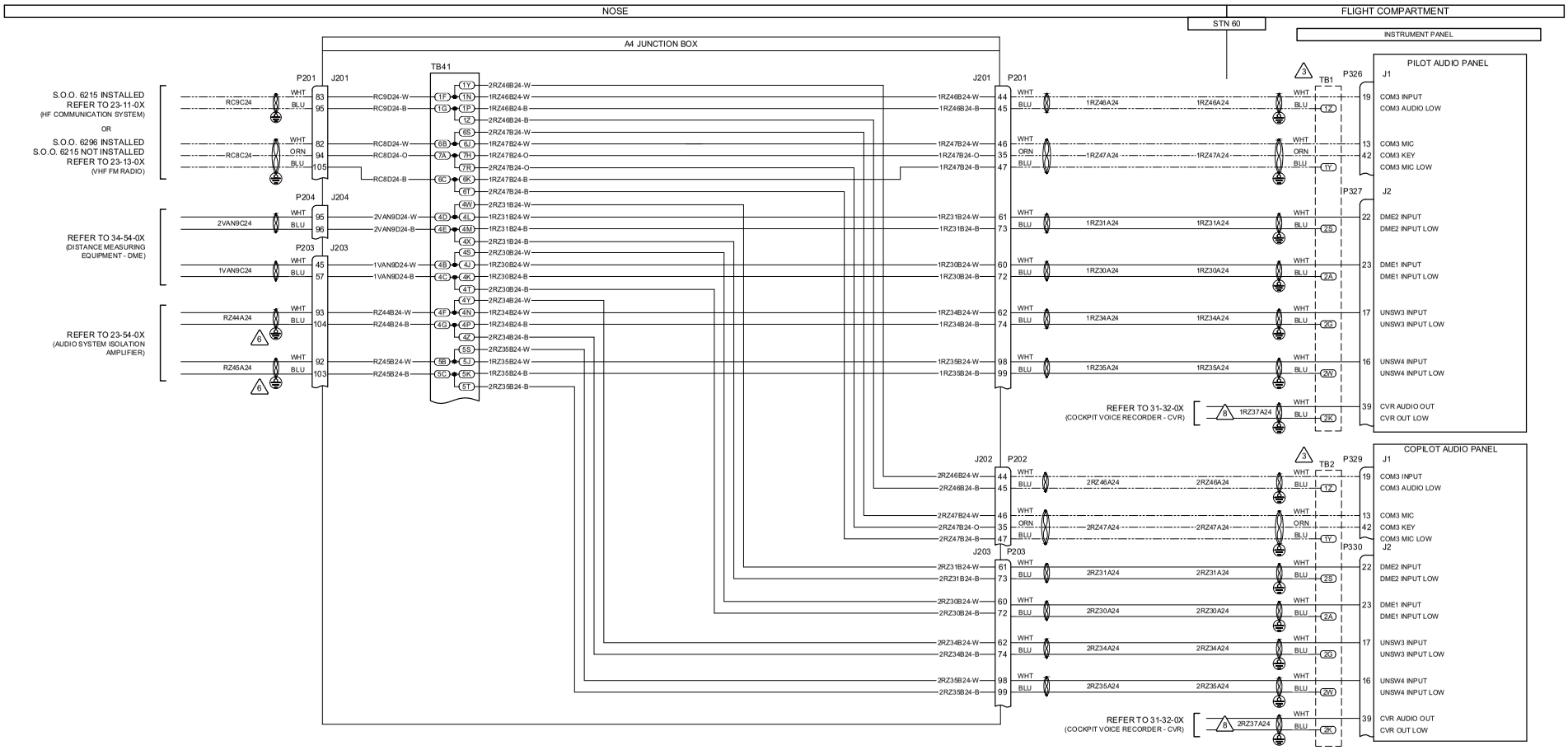


AUDIO CONTROL PANELS

SHEET 4 OF 6

C6N1004-43G ISSUE NO. 1 2 3 4

This Page Intentionally Left Blank

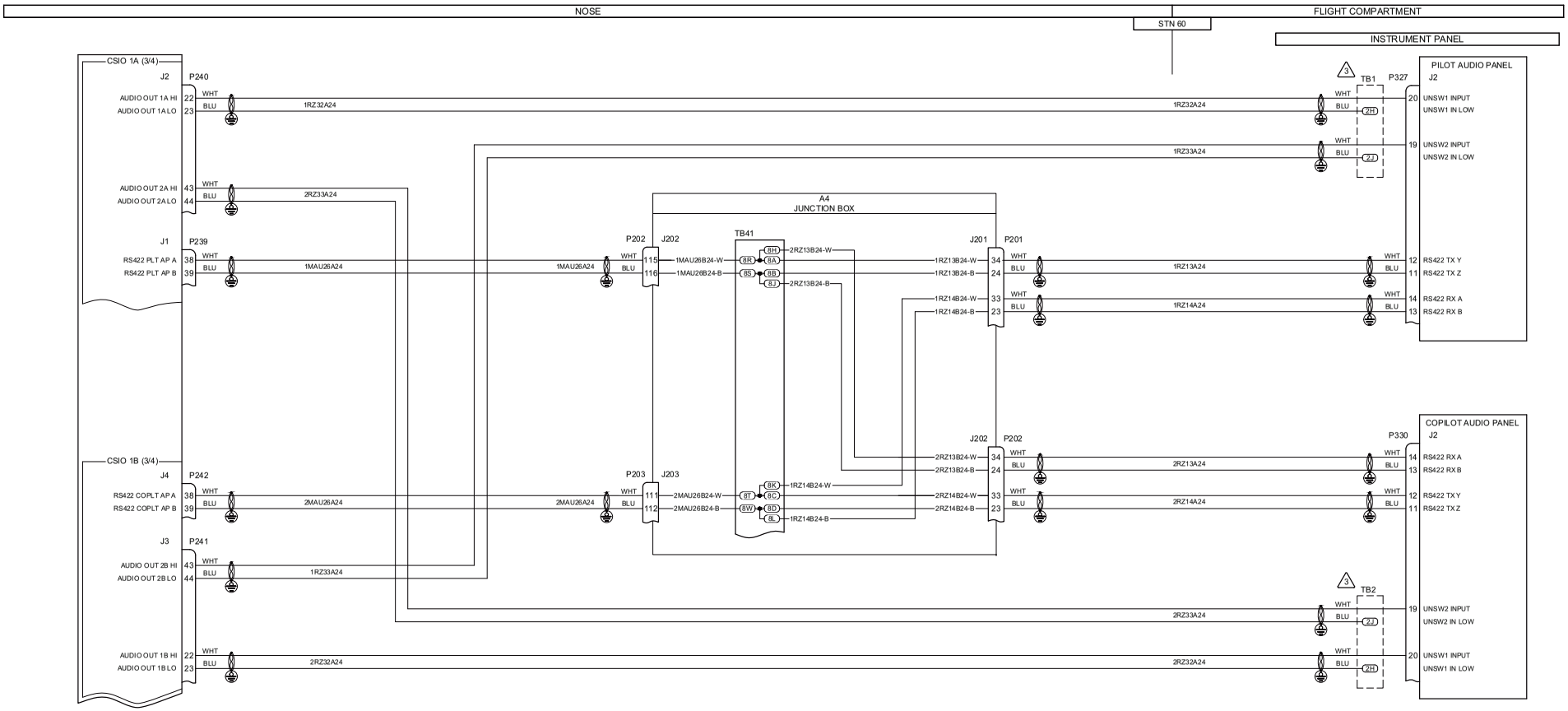


AUDIO CONTROL PANELS

SHEET 5 OF 6

C6N1004-43G ISSUE NO. 1 2 3 4

This Page Intentionally Left Blank



AUDIO CONTROL PANELS

SHEET 6 OF 6

C6N1004-43G ISSUE NO. 1 2 3 4

This Page Intentionally Left Blank

CHAPTER 23

COMMUNICATIONS

This Page Intentionally Left Blank

CHAPTER 23 – COMMUNICATIONS

This list shows the effective pages in the manual, with their latest revision date. An asterisk (*) indicates pages added or replaced at the latest revision. Technical changes to the content are indicated by a black vertical line next to the change or the paragraph that contains the change.

REFERENCE	PAGE	DATE	REFERENCE	PAGE	DATE
23	1	Sep. 30/14	23-53-00	* 1	Nov. 30/18
	2	Blank Page		* 2	Nov. 30/18
	* 3	Nov. 30/18		* 3	Nov. 30/18
	* 4	Blank Page		* 4	Nov. 30/18
	5	Sep. 30/14		* 5	Nov. 30/18
	6	Blank Page		* 6	Nov. 30/18
23-11-00	1	Sep. 30/14		* 7	Nov. 30/18
	2	Sep. 30/14		* 8	Nov. 30/18
	3	Sep. 30/14		* 9	Nov. 30/18
	4	Blank Page		* 10	Nov. 30/18
23-13-00	1	Sep. 30/14		* 11	Nov. 30/18
	2	Blank Page		* 12	Blank Page
23-16-00	1	Sep. 30/14	23-54-00	* 1	Nov. 30/18
	2	Blank Page		* 2	Blank Page

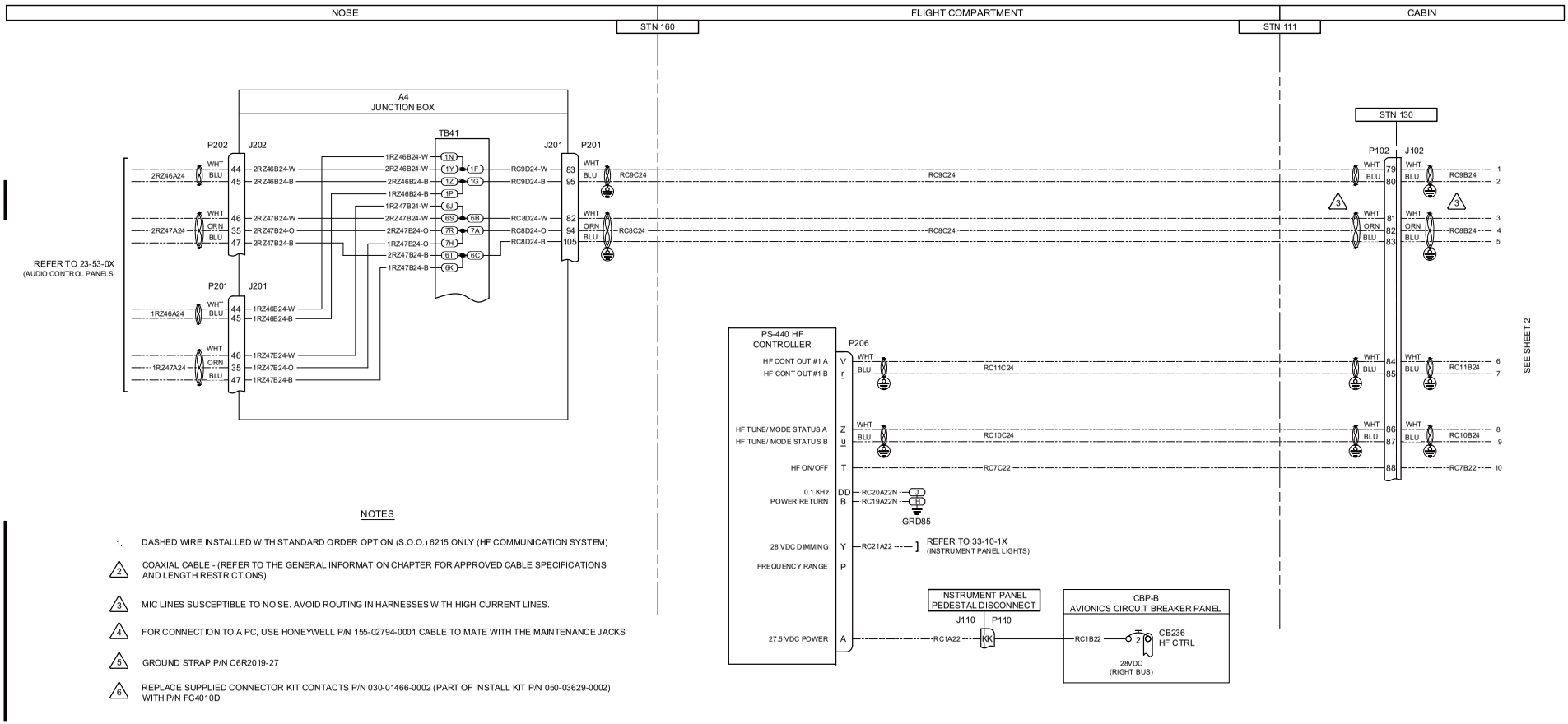
This Page Intentionally Left Blank

CHAPTER 23 – COMMUNICATIONS

TABLE OF CONTENTS

SUBJECT	CHAPTER SECTION SUBJECT
SPEECH COMMUNICATIONS	**23-10**
– HF COMMUNICATION	**23-11**
HF Communication System (S.O.O. 6215)	23-11-00
– VHF COMMUNICATION	**23-13**
VHF-FM Radio (S.O.O. 6296)	23-13-00
– ECTM/FOQA DATA CAPTURE & SATELLITE COMMUNICATION SYSTEM	**23-15**
ECTM-FOQA & SAT COM (S.O.O. 6288 & S.O.O. 6298)	23-15-00
– XM SATELLITE	**23-16**
XM Satellite Weather System (S.O.O. 6211)	23-16-00
AUDIO INTEGRATING	**23-50**
– CONTROL PANELS	**23-53**
Audio Control Panels	23-53-00
– ISOLATION AMPLIFIER	**23-54**
Audio System Isolation Amplifier	23-54-00

This Page Intentionally Left Blank



NOTES

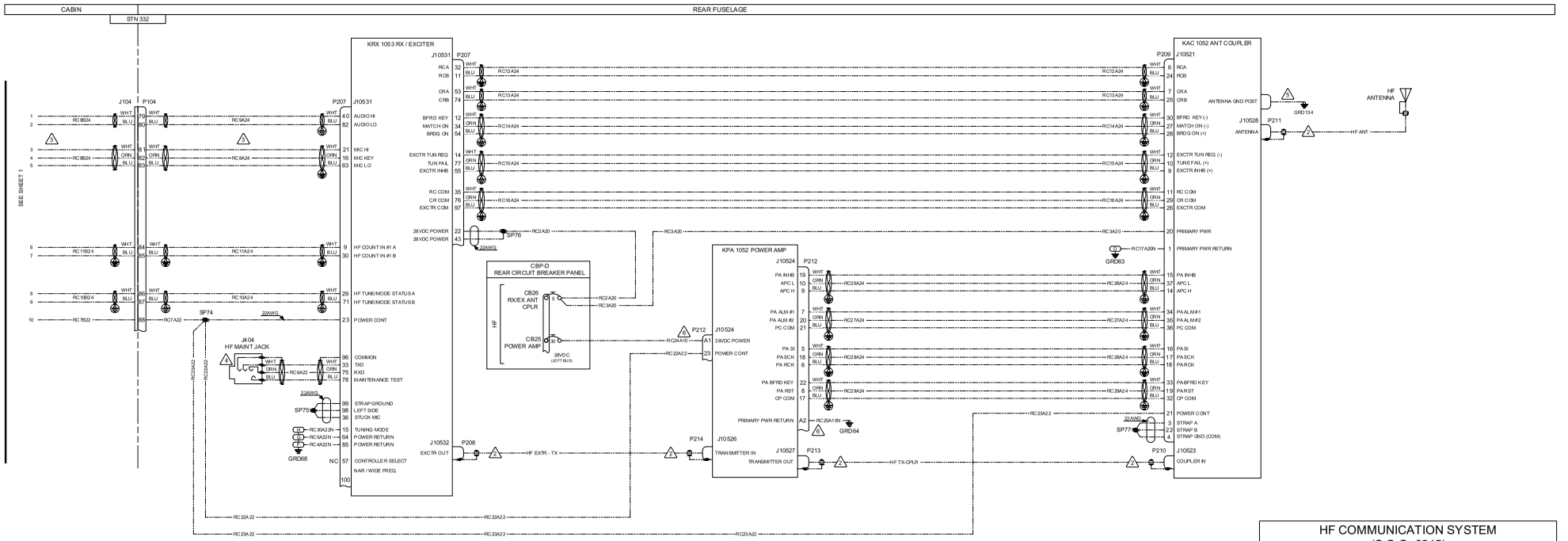
1. DASHED WIRE INSTALLED WITH STANDARD ORDER OPTION (S.O.O.) 6215 ONLY (HF COMMUNICATION SYSTEM)
2. COAXIAL CABLE - (REFER TO THE GENERAL INFORMATION CHAPTER FOR APPROVED CABLE SPECIFICATIONS AND LENGTH RESTRICTIONS)
3. MIC LINES SUSCEPTIBLE TO NOISE. AVOID ROUTING IN HARNESSES WITH HIGH CURRENT LINES.
4. FOR CONNECTION TO A PC, USE HONEYWELL P/N 155-02794-0001 CABLE TO MATE WITH THE MAINTENANCE JACKS
5. GROUND STRAP P/N C6R2019-27
6. REPLACE SUPPLIED CONNECTOR KIT CONTACTS P/N 030-01466-0002 (PART OF INSTALL KIT P/N 050-03629-0002) WITH P/N FC4010D

HF COMMUNICATION SYSTEM
(S.O.O. 6215)

SHEET 1 OF 2

C6N1004-41D	ISSUE NO.	1	2						

This Page Intentionally Left Blank

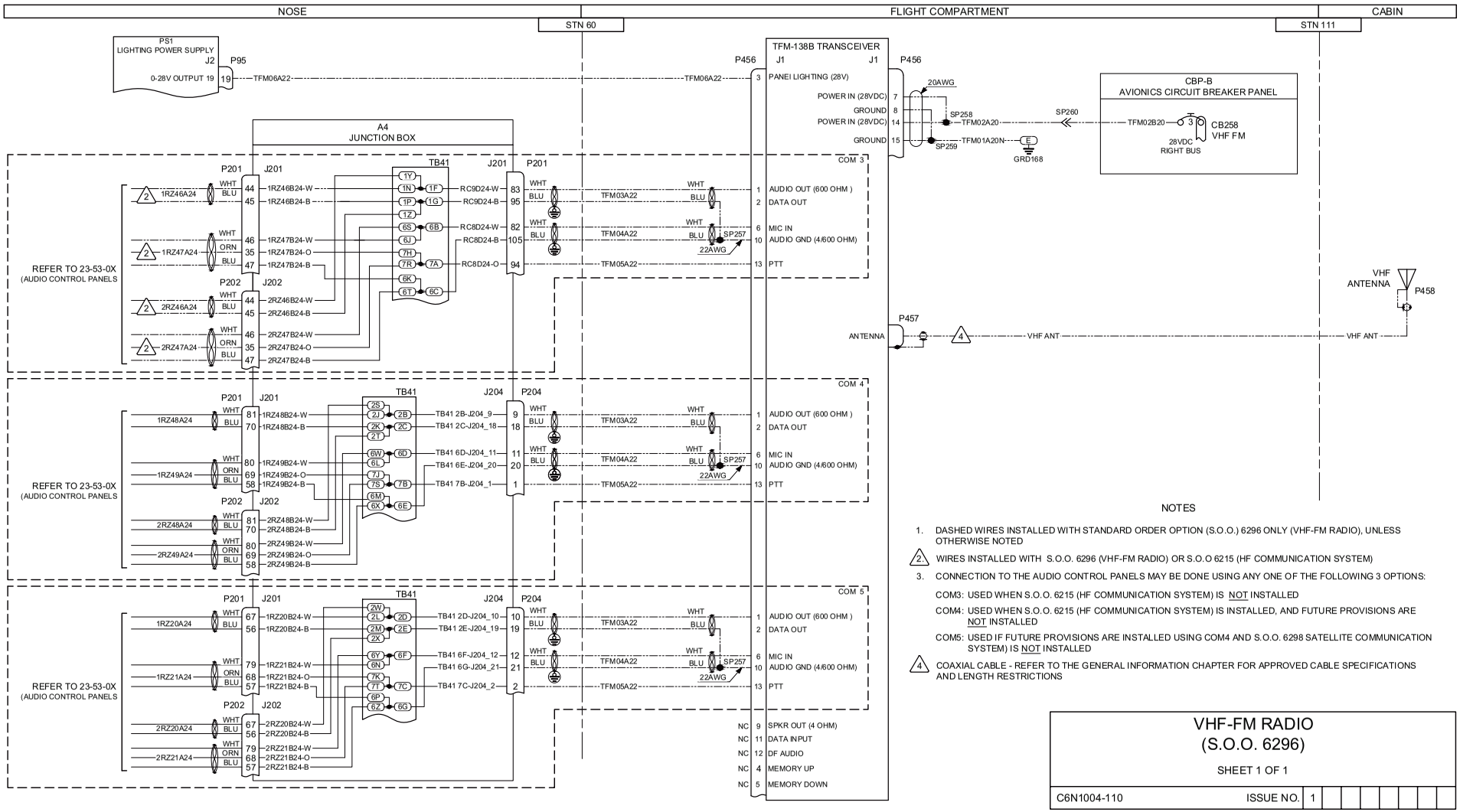


HF COMMUNICATION SYSTEM
(S.O.O. 6215)

SHEET 2 OF 2

C6N1004-41D ISSUE NO. 1 2

This Page Intentionally Left Blank



NOTES

1. DASHED WIRES INSTALLED WITH STANDARD ORDER OPTION (S.O.O.) 6296 ONLY (VHF-FM RADIO), UNLESS OTHERWISE NOTED
2. WIRES INSTALLED WITH S.O.O. 6296 (VHF-FM RADIO) OR S.O.O 6215 (HF COMMUNICATION SYSTEM)
3. CONNECTION TO THE AUDIO CONTROL PANELS MAY BE DONE USING ANY ONE OF THE FOLLOWING 3 OPTIONS:

   COM3: USED WHEN S.O.O. 6215 (HF COMMUNICATION SYSTEM) IS NOT INSTALLED

   COM4: USED WHEN S.O.O. 6215 (HF COMMUNICATION SYSTEM) IS INSTALLED, AND FUTURE PROVISIONS ARE NOT INSTALLED

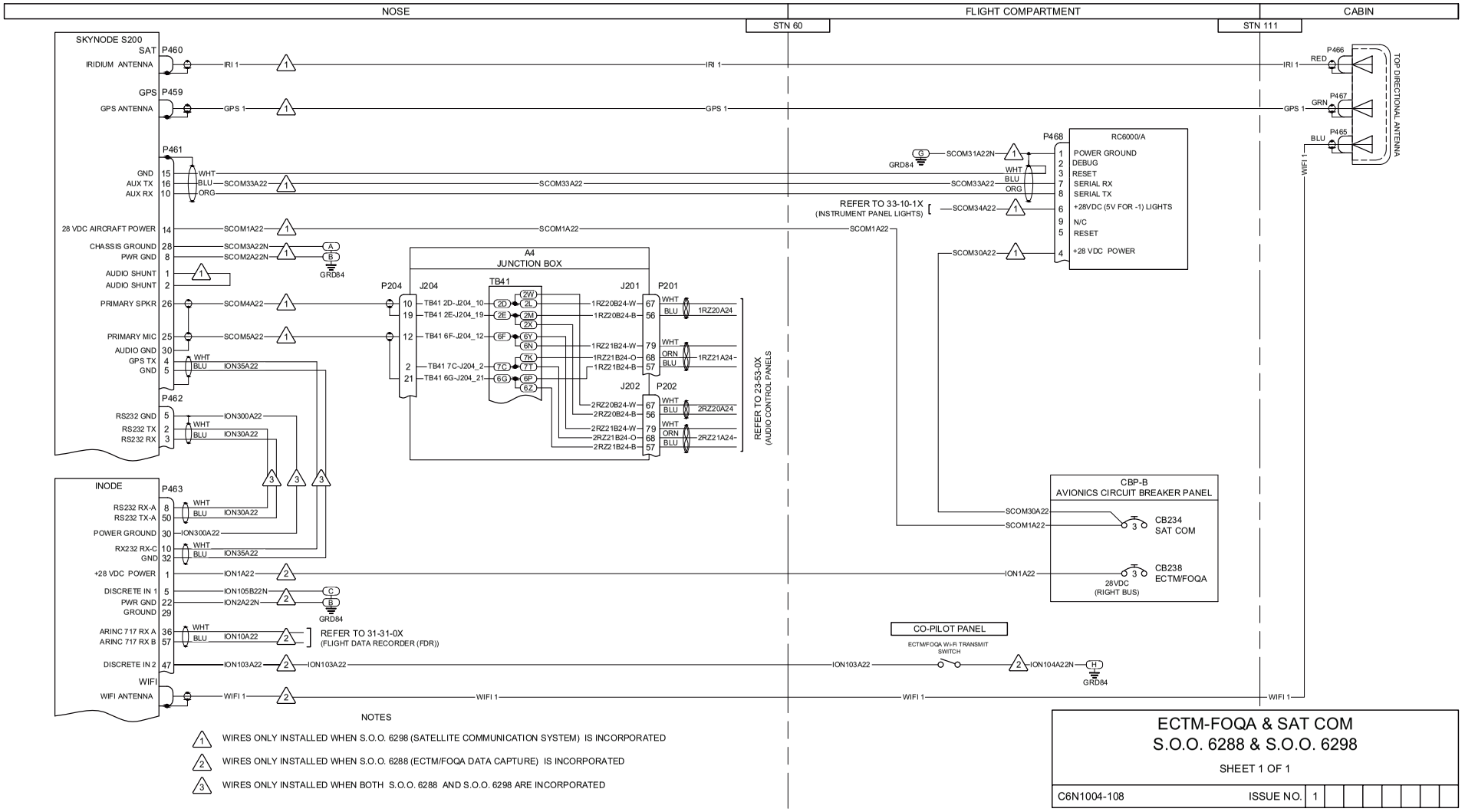
   COM5: USED IF FUTURE PROVISIONS ARE INSTALLED USING COM4 AND S.O.O. 6298 SATELLITE COMMUNICATION SYSTEM) IS NOT INSTALLED
4. COAXIAL CABLE - REFER TO THE GENERAL INFORMATION CHAPTER FOR APPROVED CABLE SPECIFICATIONS AND LENGTH RESTRICTIONS

VHF-FM RADIO
(S.O.O. 6296)

SHEET 1 OF 1

C6N1004-110	ISSUE NO.	1

This Page Intentionally Left Blank



NOTES

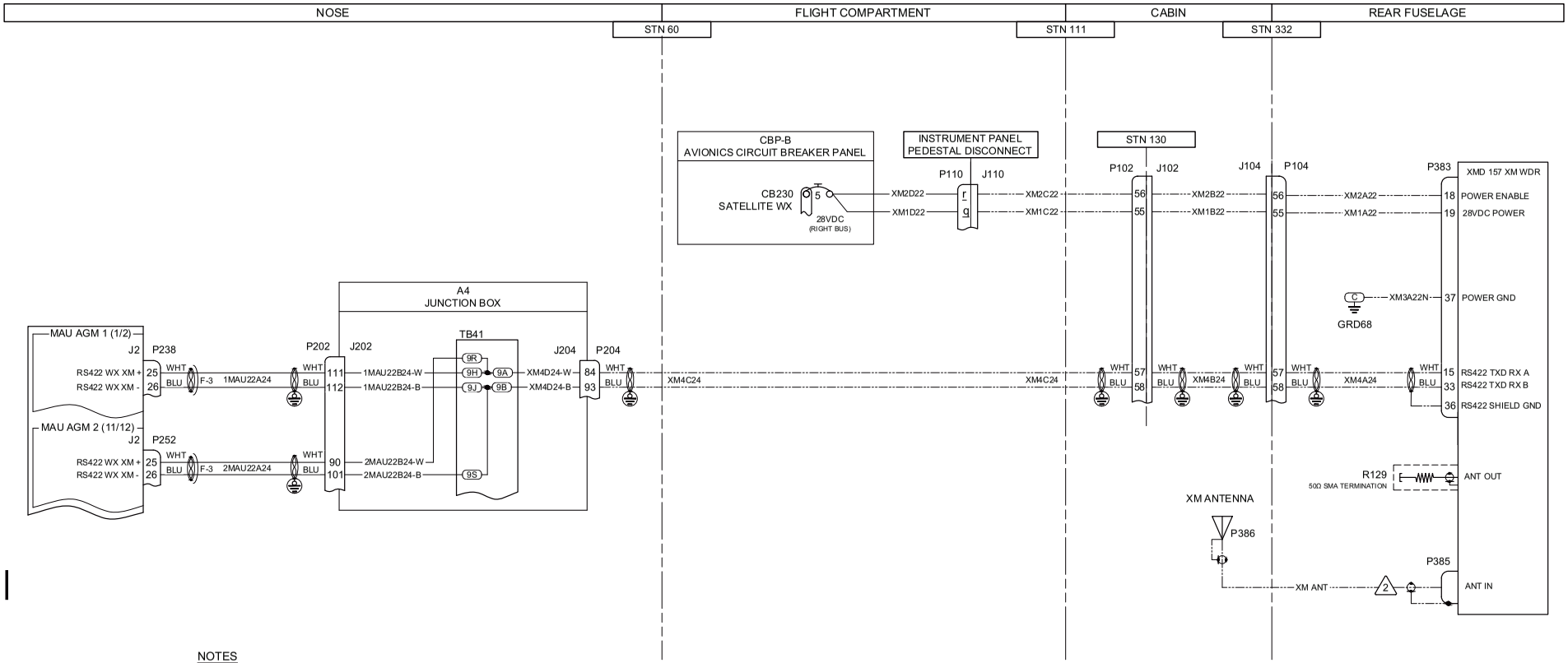
1. WIRES ONLY INSTALLED WHEN S.O.O. 6298 (SATELLITE COMMUNICATION SYSTEM) IS INCORPORATED
2. WIRES ONLY INSTALLED WHEN S.O.O. 6288 (ECTM/FOQA DATA CAPTURE) IS INCORPORATED
3. WIRES ONLY INSTALLED WHEN BOTH S.O.O. 6288 AND S.O.O. 6298 ARE INCORPORATED

ECTM-FOQA & SAT COM
S.O.O. 6288 & S.O.O. 6298

SHEET 1 OF 1

C6N1004-108 ISSUE NO. 1

This Page Intentionally Left Blank



NOTES

1. DASHED WIRE INSTALLED WITH STANDARD ORDER OPTION (S.O.O.) 6211 ONLY (XM SATELLITE WEATHER SYSTEM)

2 COAXIAL CABLE - (REFER TO THE GENERAL INFORMATION CHAPTER FOR APPROVED CABLE SPECIFICATIONS AND LENGTH RESTRICTIONS)

XM SATELLITE WEATHER SYSTEM (S.O.O. 6211)

SHEET 1 OF 1

C6N1004-42D	ISSUE NO.	1	2						

This Page Intentionally Left Blank